



US012414495B2

(12) **United States Patent**
Radtke

(10) **Patent No.:** **US 12,414,495 B2**

(45) **Date of Patent:** **Sep. 16, 2025**

(54) **SCRAPER FOR TRENCH CLOSING
ASSEMBLY WHEEL**

(71) Applicant: **Precision Planting LLC**, Tremont, IL
(US)

(72) Inventor: **Ian Radtke**, Washington, IL (US)

(73) Assignee: **Precision Planting LLC**, Tremont, IL
(US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 841 days.

(21) Appl. No.: **17/615,761**

(22) PCT Filed: **Jun. 17, 2020**

(86) PCT No.: **PCT/IB2020/055638**

§ 371 (c)(1),

(2) Date: **Dec. 1, 2021**

(87) PCT Pub. No.: **WO2020/261043**

PCT Pub. Date: **Dec. 30, 2020**

(65) **Prior Publication Data**

US 2022/0304219 A1 Sep. 29, 2022

Related U.S. Application Data

(60) Provisional application No. 62/866,700, filed on Jun.
26, 2019.

(51) **Int. Cl.**
A01C 5/06 (2006.01)

(52) **U.S. Cl.**
CPC **A01C 5/068** (2013.01)

(58) **Field of Classification Search**

CPC A01C 23/023–028; A01C 5/06–068; A01C
7/06

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,141,302 A 2/1979 Morrison, Jr. et al.
4,878,443 A * 11/1989 Gardner A01C 5/064
172/538

5,443,125 A 8/1995 Clark et al.
5,626,196 A 5/1997 Hughes
5,704,431 A 1/1998 Bercheid

(Continued)

FOREIGN PATENT DOCUMENTS

AU 24551 84 A 9/1984
CN 201557379 U 8/2010

(Continued)

OTHER PUBLICATIONS

UK Intellectual Property Office, Search Report for related UK
Patent Application No. GB1910423.I, dated Jan. 17, 2020.

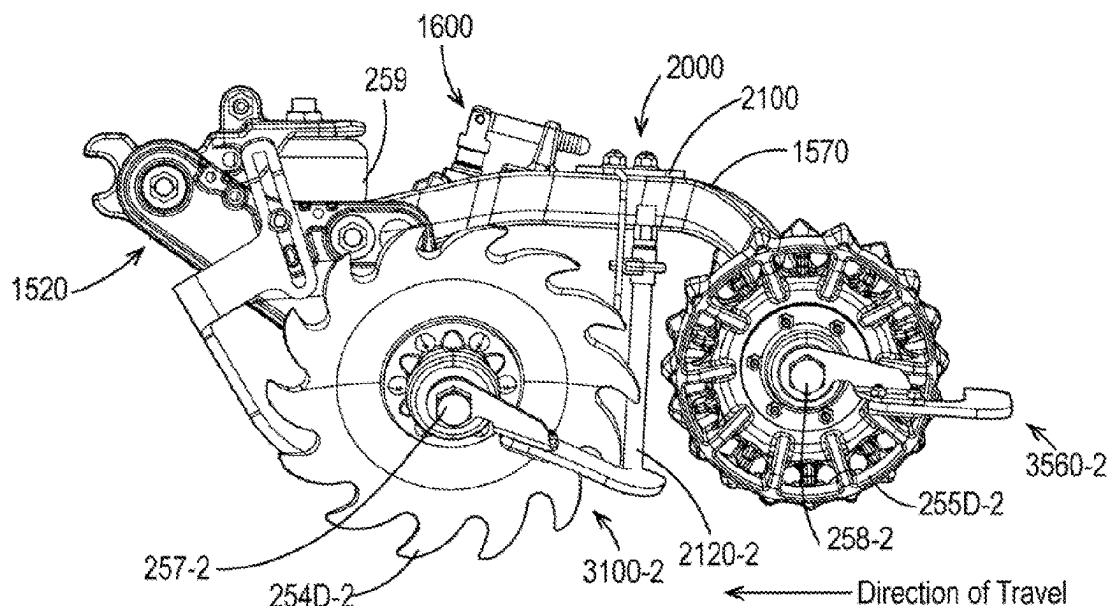
(Continued)

Primary Examiner — Tara Mayo

(57) **ABSTRACT**

A trench closing assembly (250) for a row unit (200) of an
agricultural planter, comprising a main frame (251) and a
frame member (1520) pivotally supported from the main
frame, a pair of closing wheels (254) each rotatably sup-
ported by the frame member, a mounting arm (1570) piv-
otally mounted to said frame member and supporting a press
wheel (255) rearward of said closing wheels, including a
scraper (3600, 4600) adjacent to an inner side of said press
wheel.

9 Claims, 28 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

6,530,334	B2	3/2003	Hagny	
7,322,302	B1	1/2008	Reidhar	
8,561,472	B2	10/2013	Sauder et al.	
9,173,339	B2	11/2015	Sauder et al.	
9,872,423	B2	1/2018	Anderson et al.	
2009/0056962	A1	3/2009	Martin et al.	
2013/0206431	A1	8/2013	Freed	
2017/0000021	A1 *	1/2017	Schaffert et al.	 A01B 25/00
2017/0231149	A1	8/2017	Devloo et al.	

FOREIGN PATENT DOCUMENTS

CN	206851304	U	1/2018
CN	206865998	U	1/2018
WO	2014/186810		11/2014
WO	2017/197274	A1	11/2017
WO	2019/169369	A1	9/2019
WO	2020/208442	A1	10/2020
WO	2020/225623	A1	11/2020

OTHER PUBLICATIONS

International Search Report prepared for related International Patent
Application No. PCT/1B2020/055638, mail date Sep. 17, 2020.

* cited by examiner

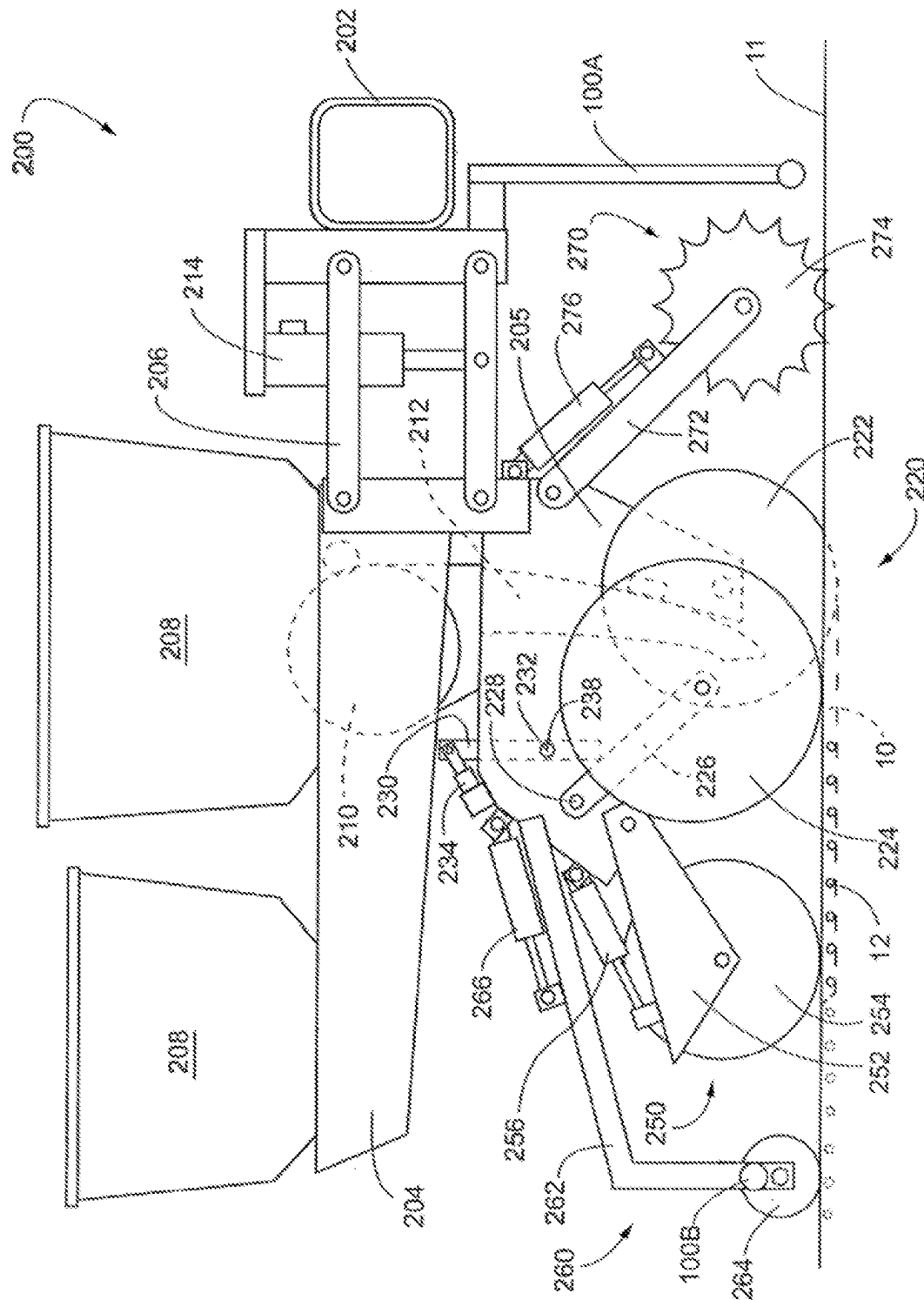


FIG. 1

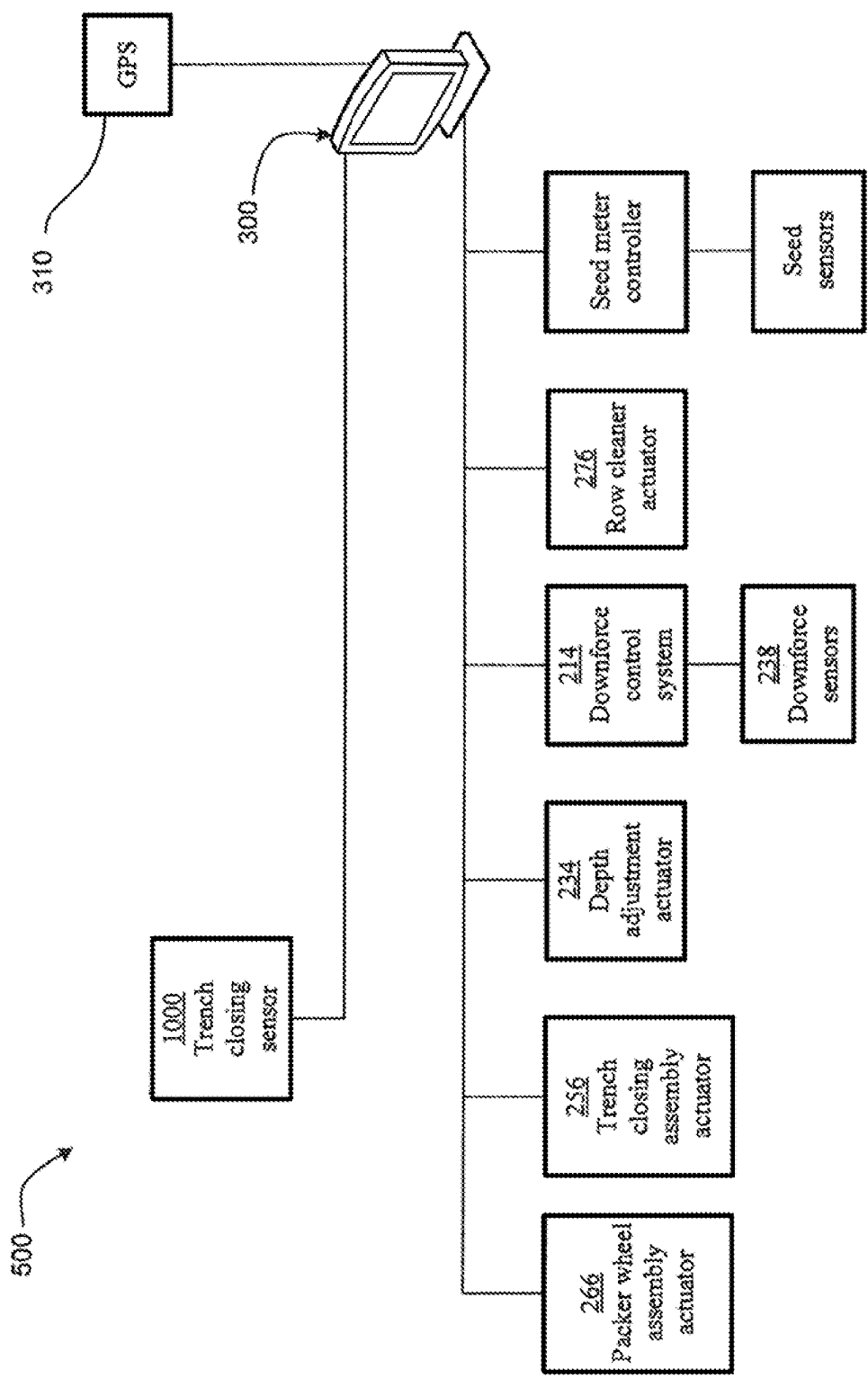


FIG. 2

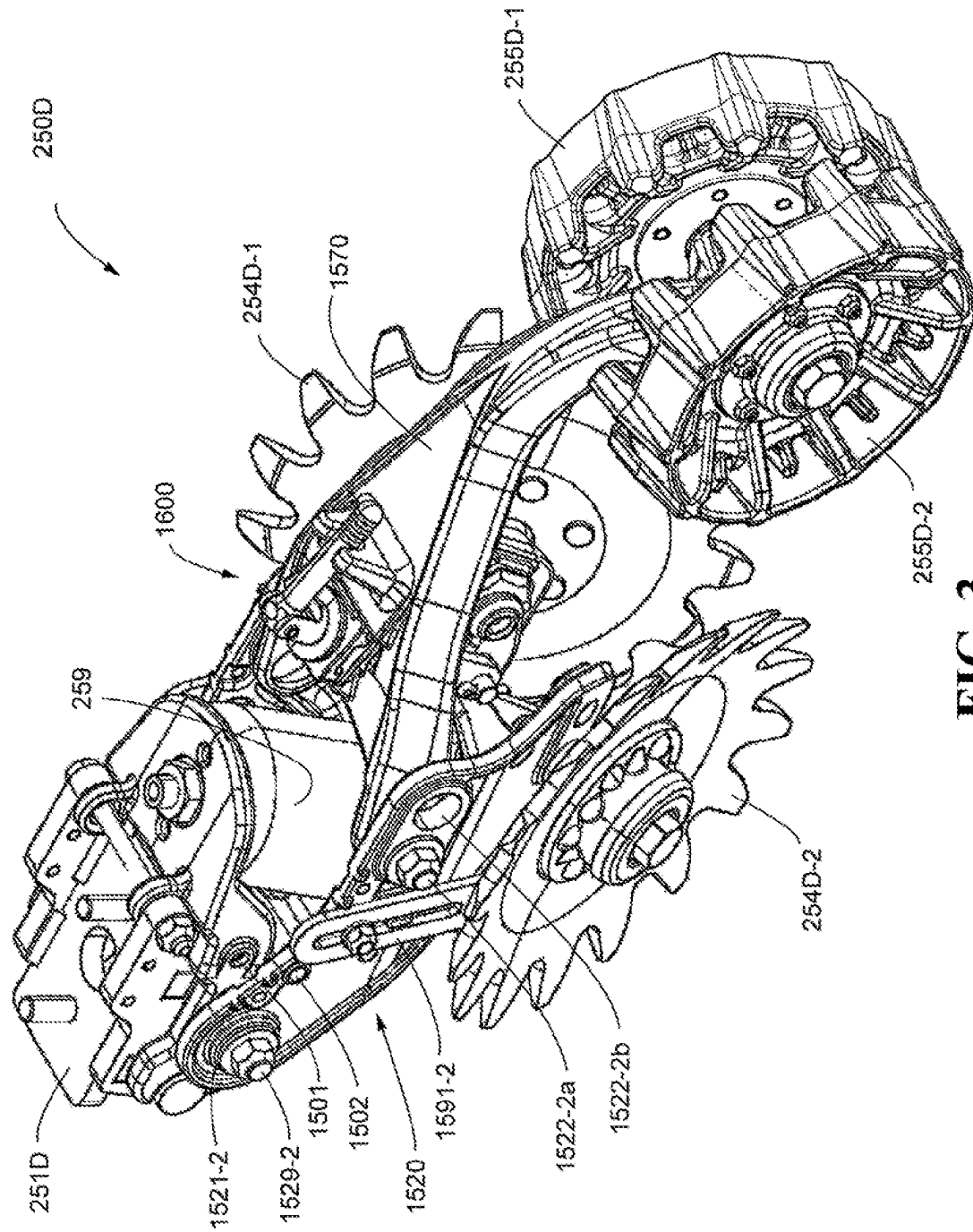


FIG. 3

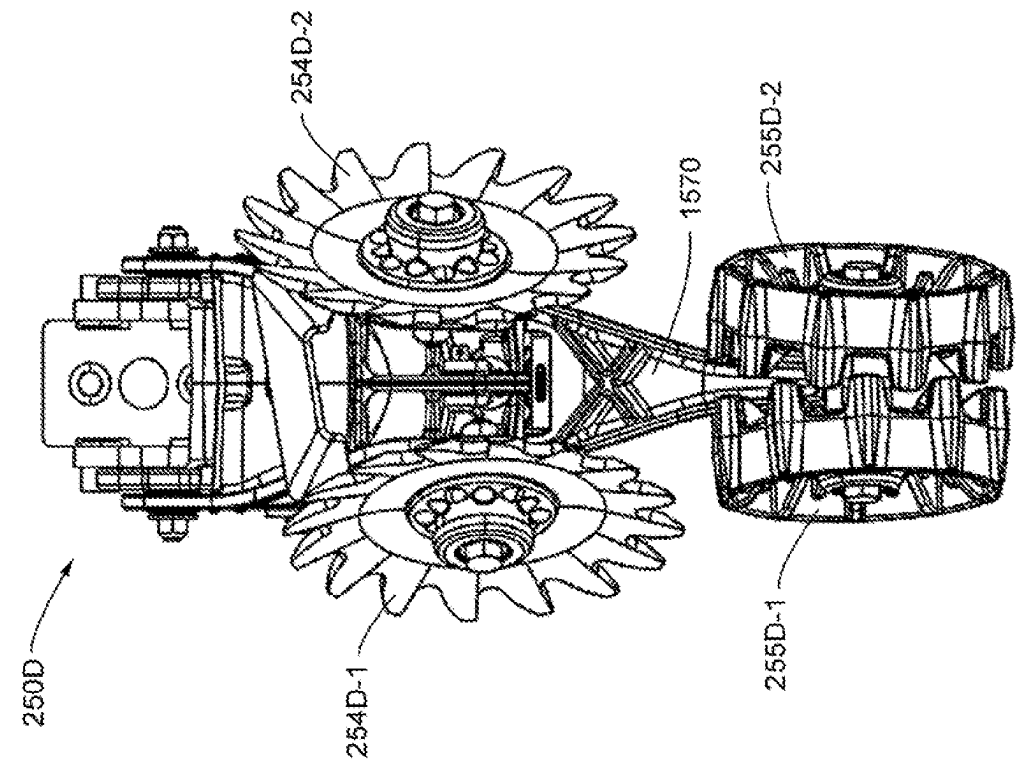


FIG. 5

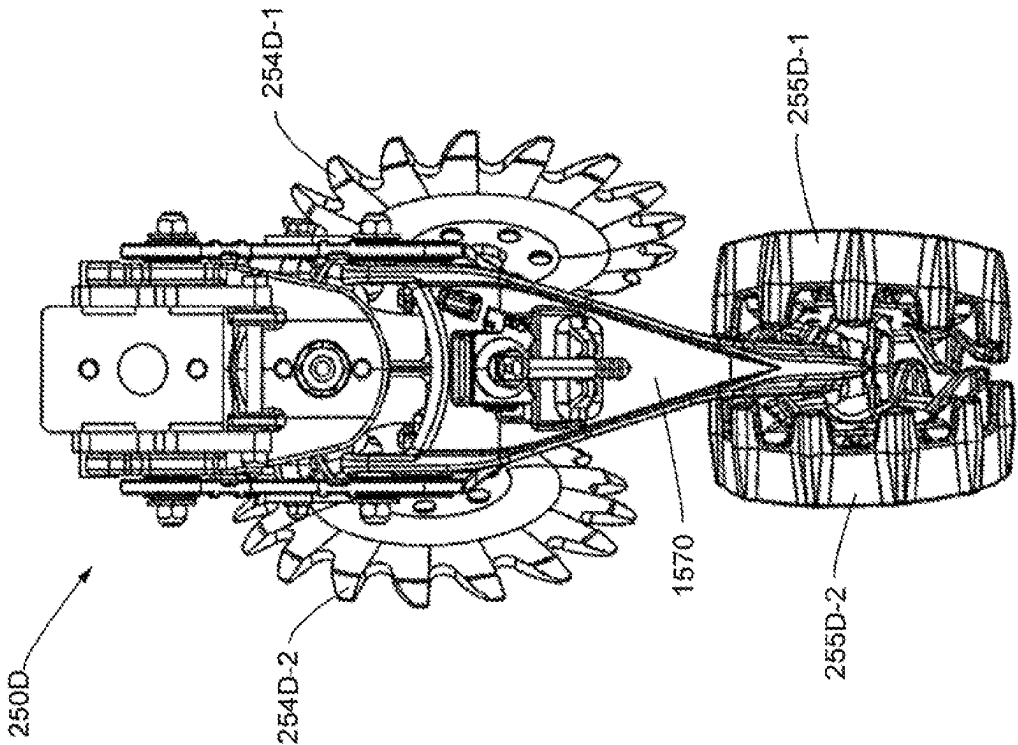


FIG. 4

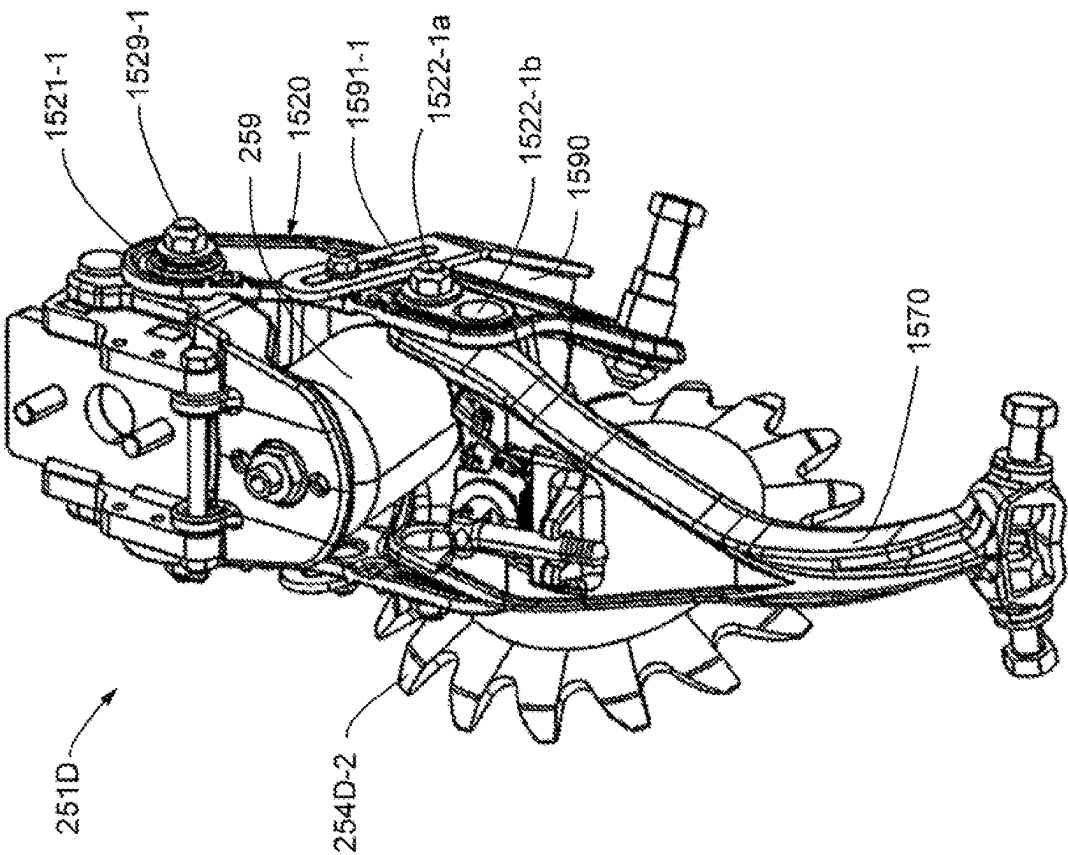


FIG. 6

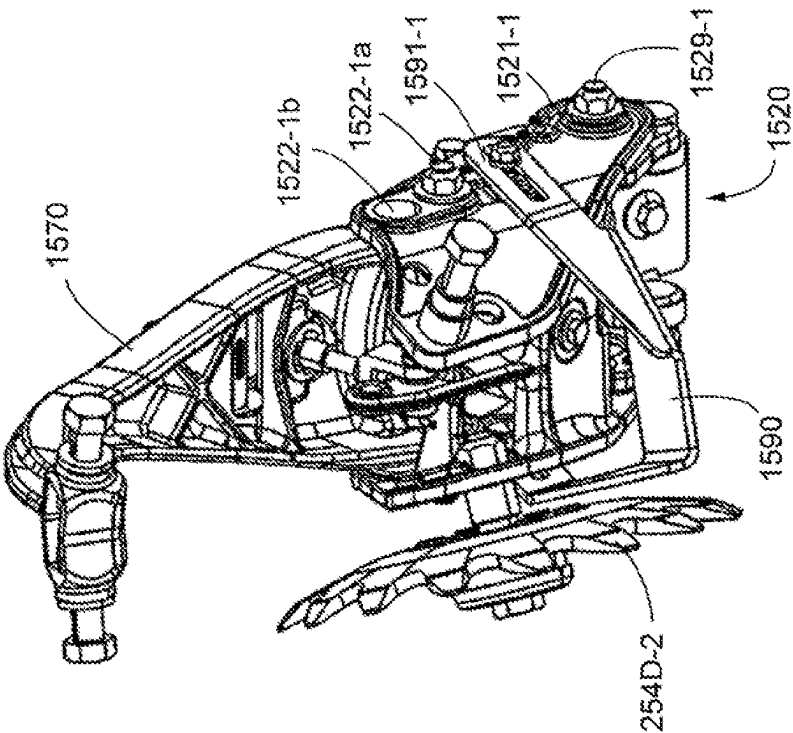


FIG. 7

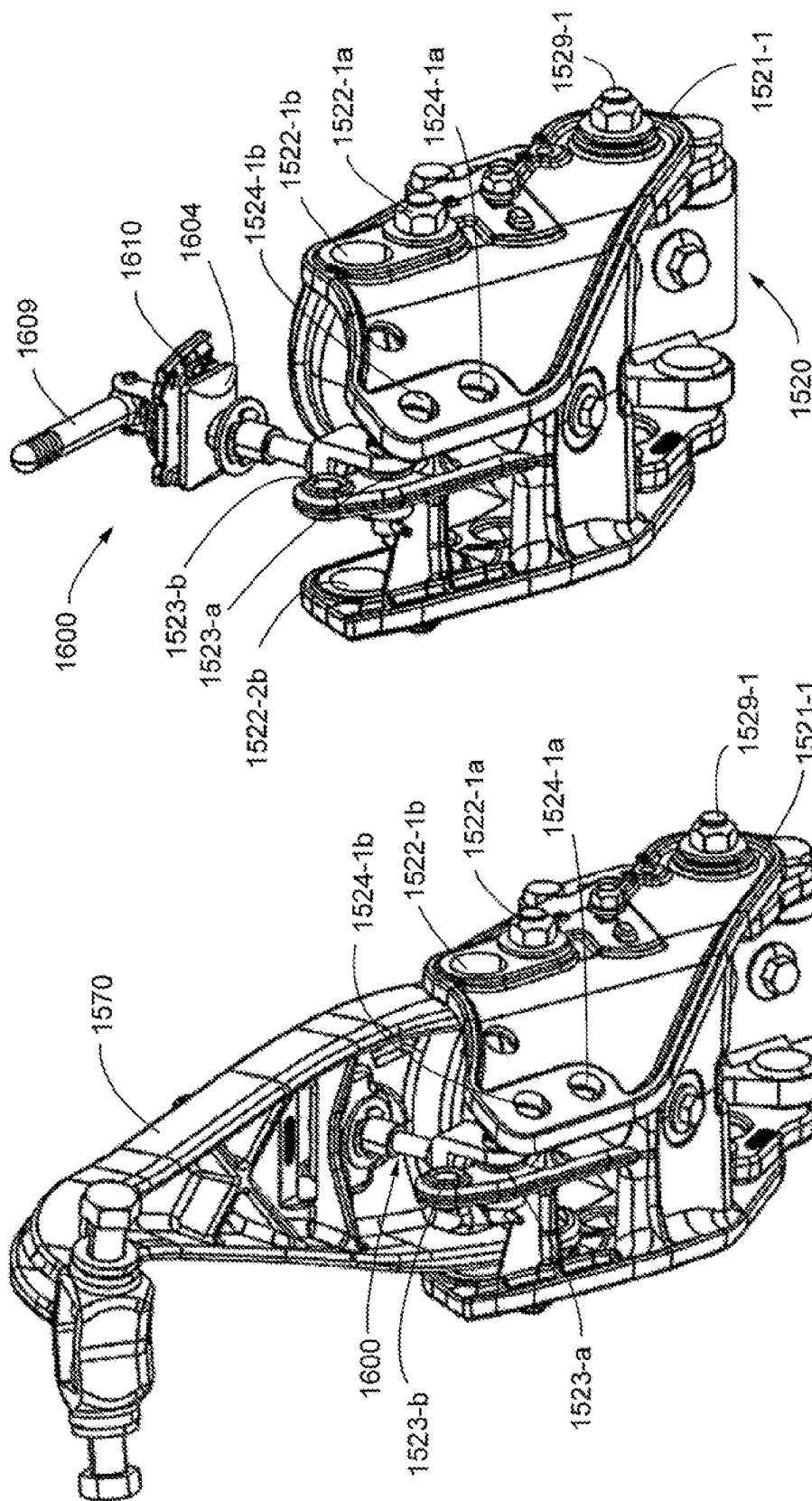


FIG. 9

FIG. 8

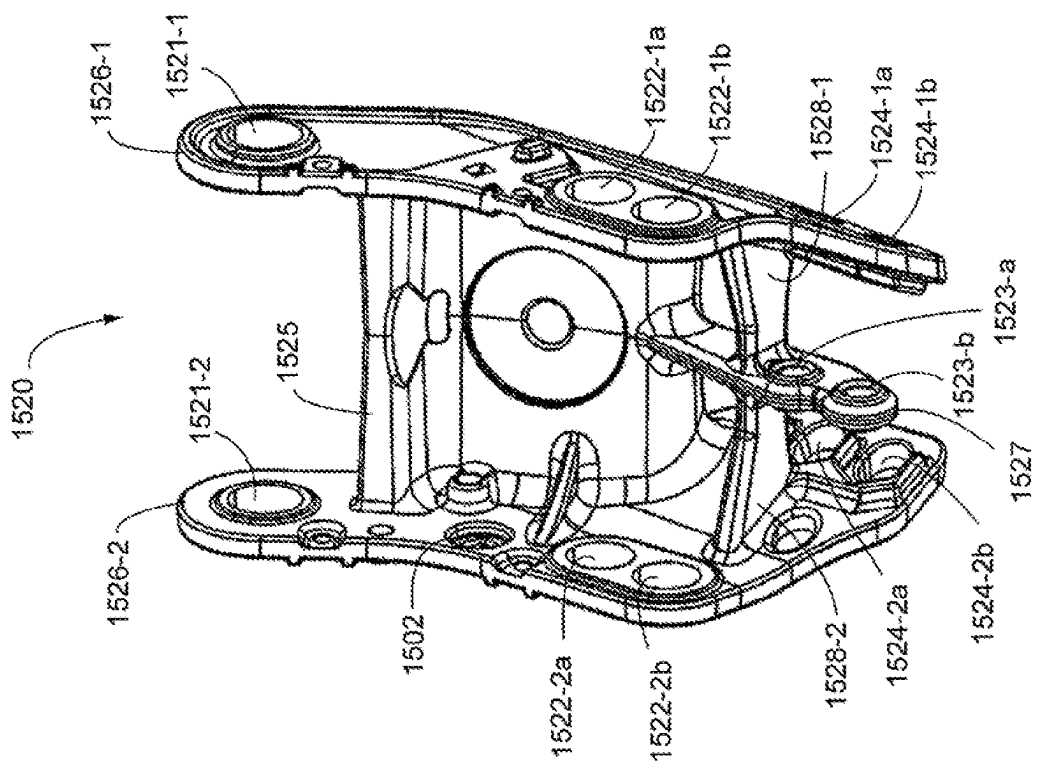


FIG. 11

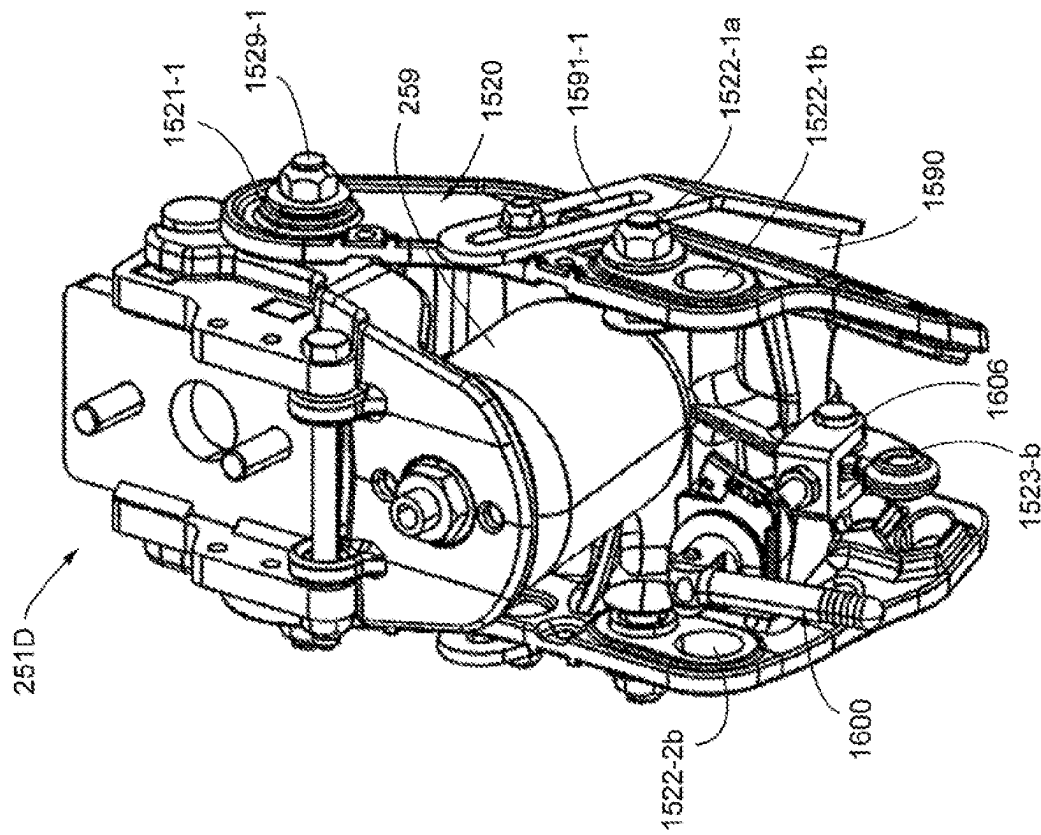


FIG. 10

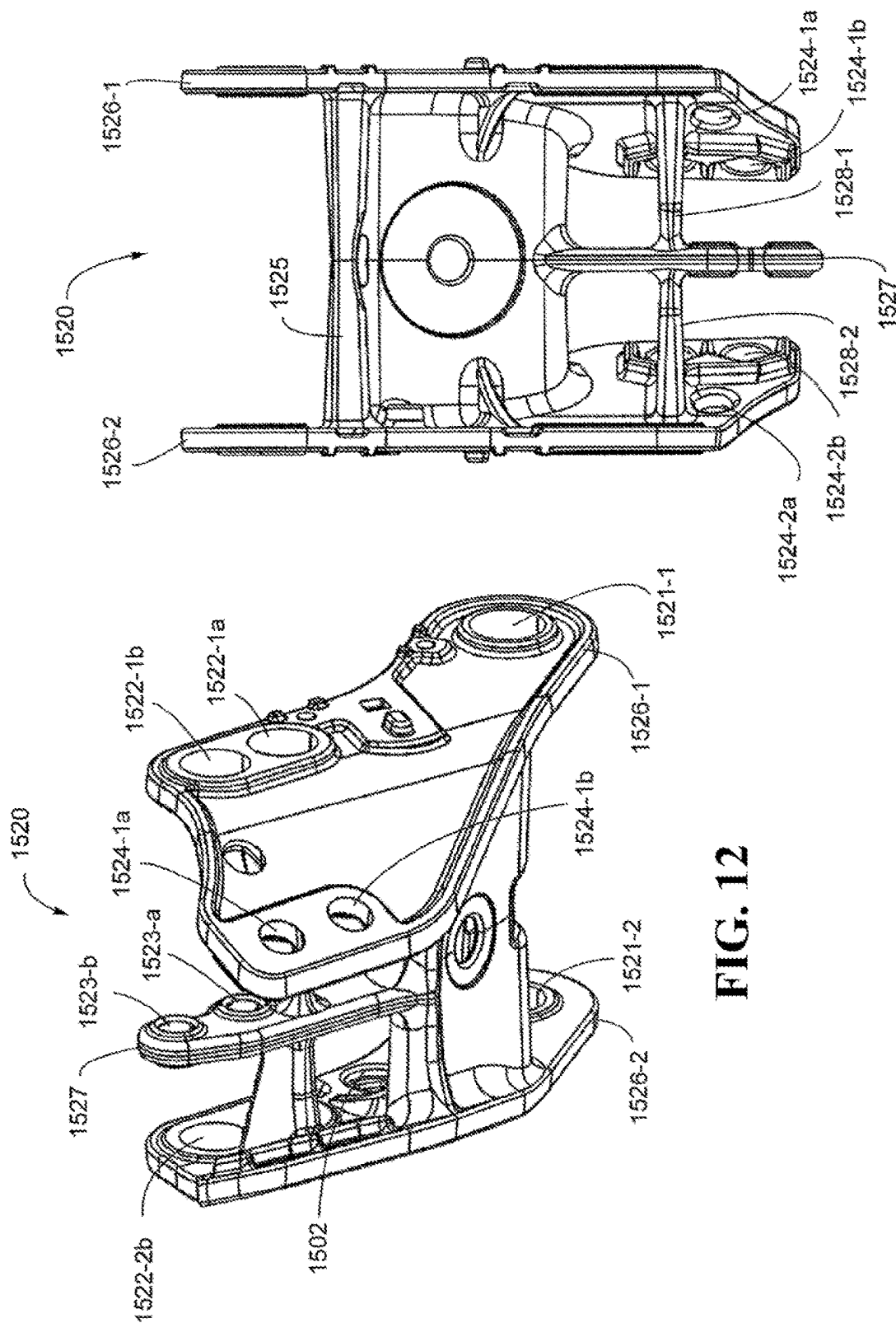


FIG. 13

FIG. 12

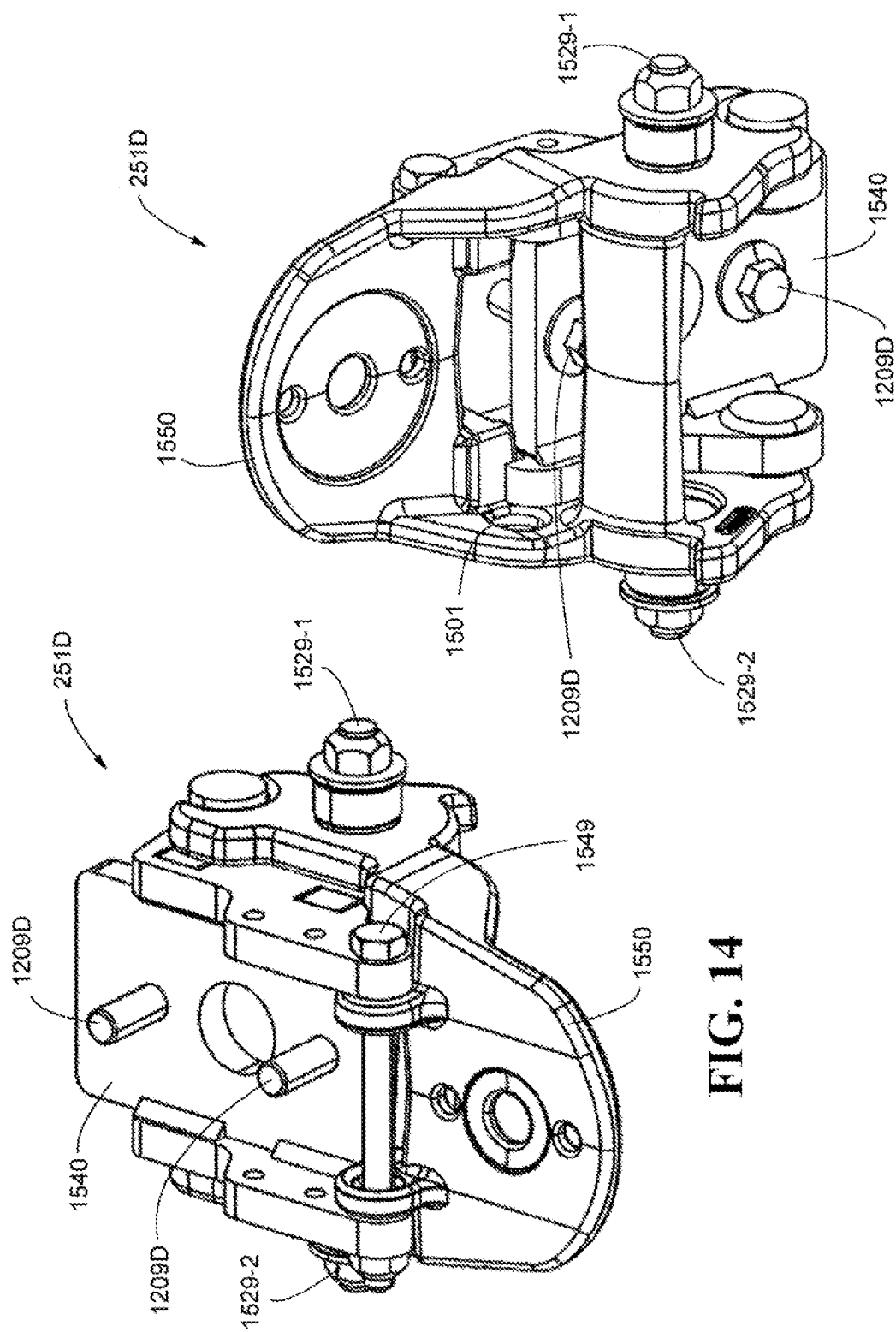


FIG. 14

FIG. 15

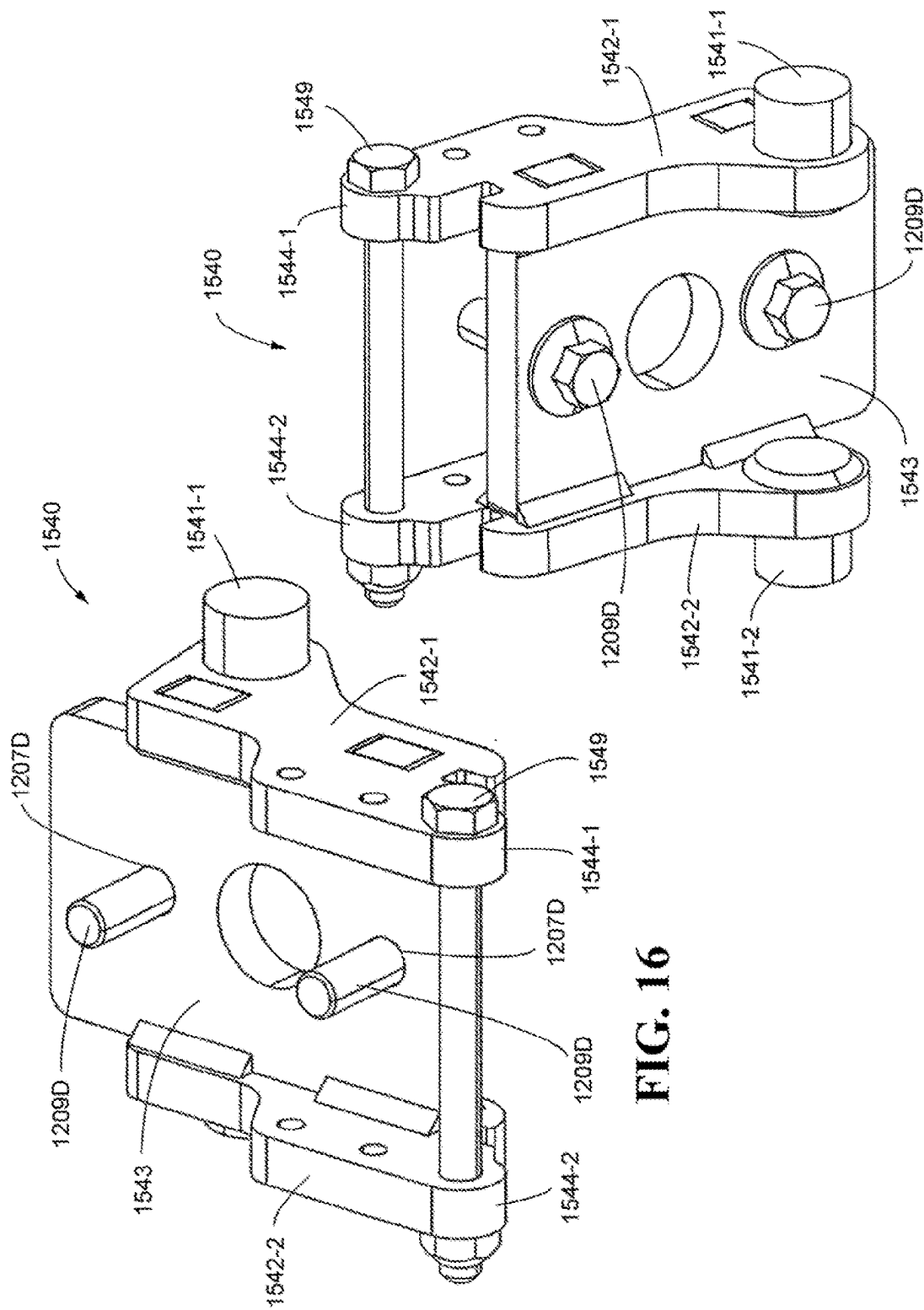
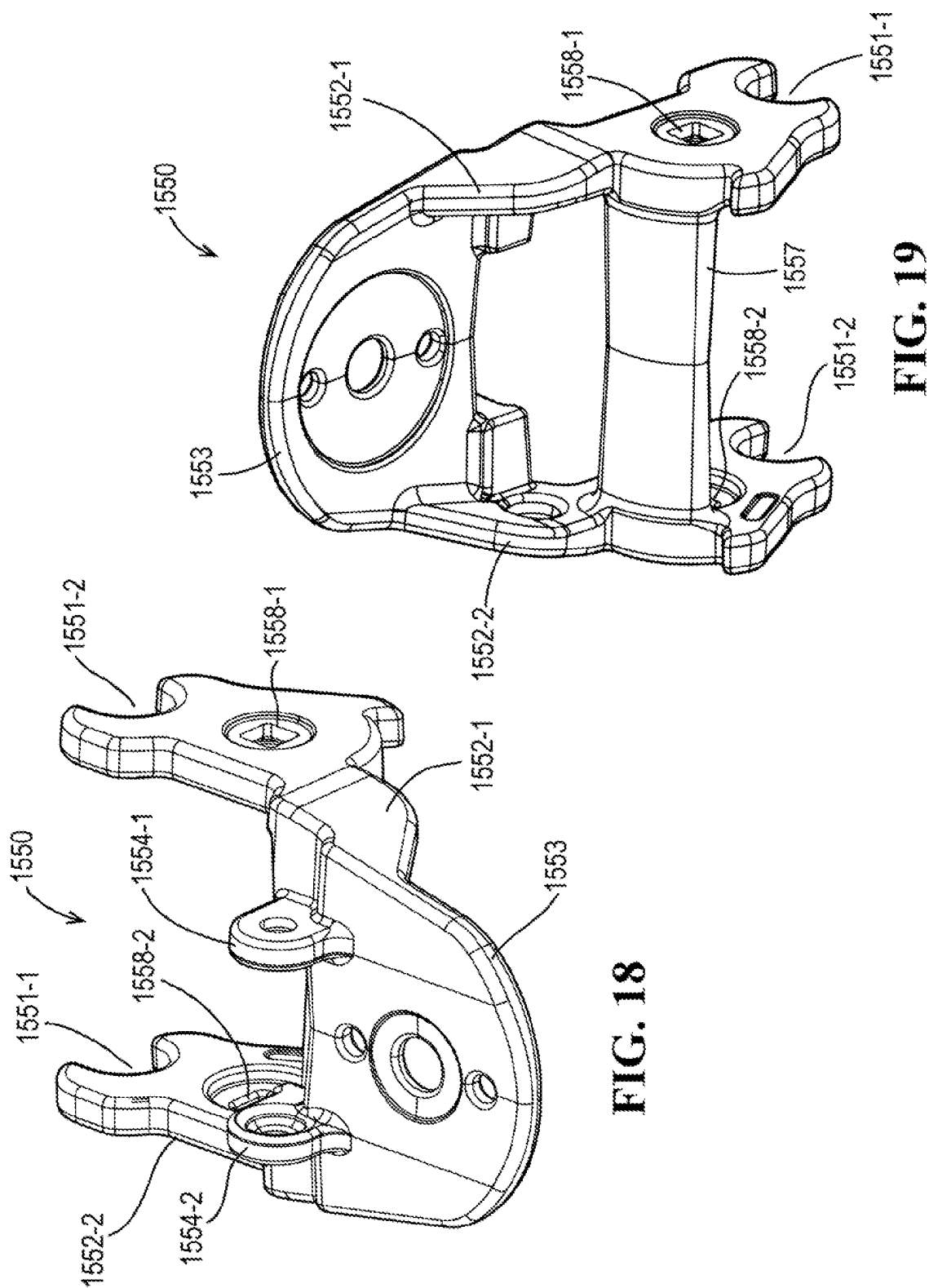


FIG. 17

FIG. 16



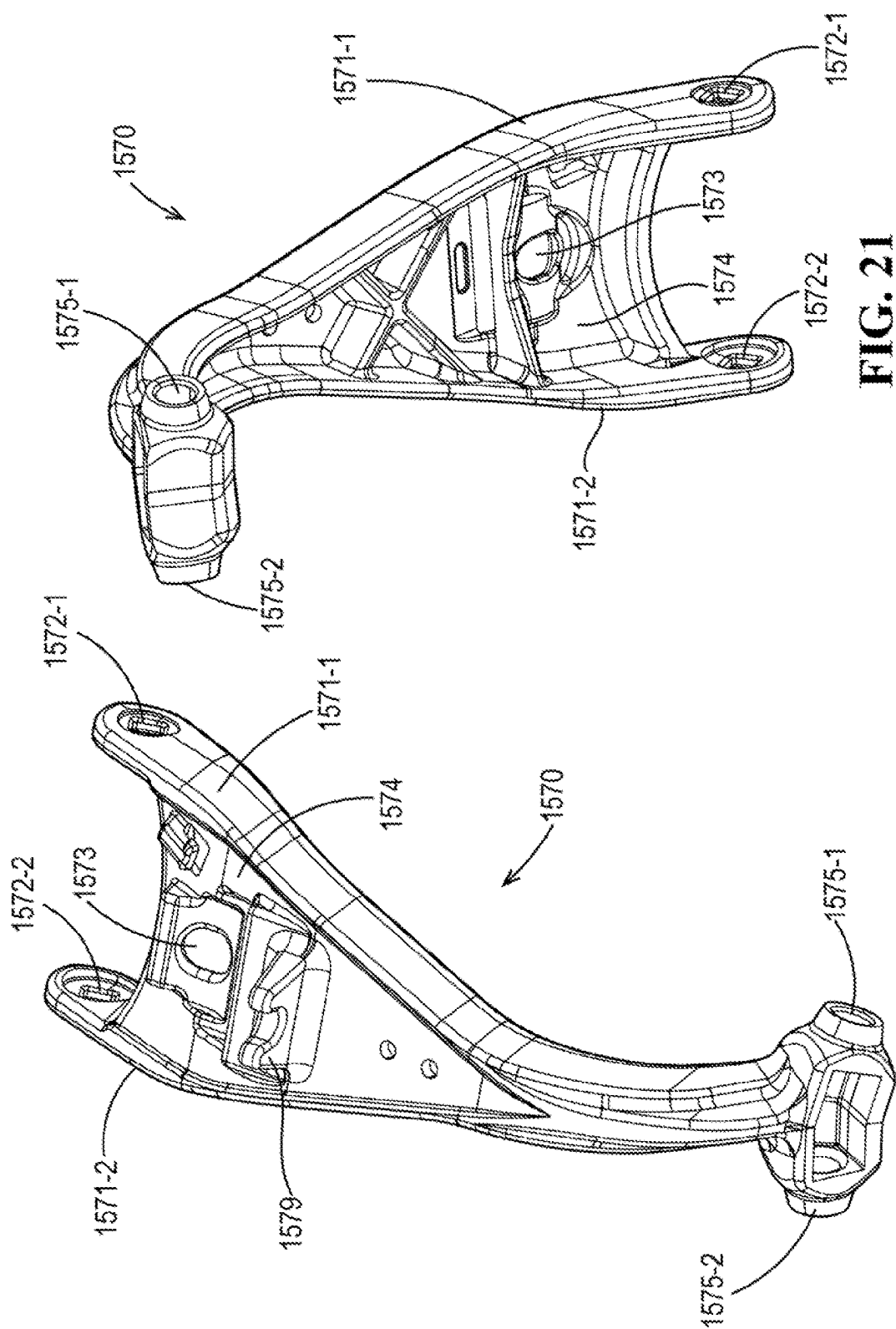


FIG. 20

FIG. 21

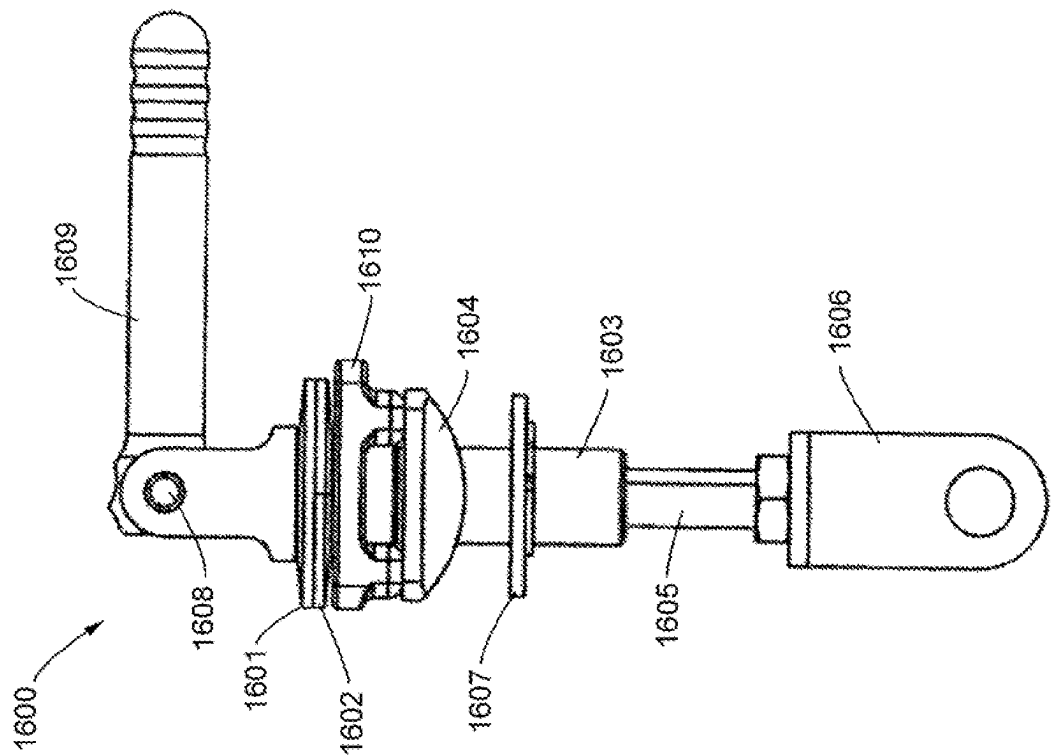


FIG. 23

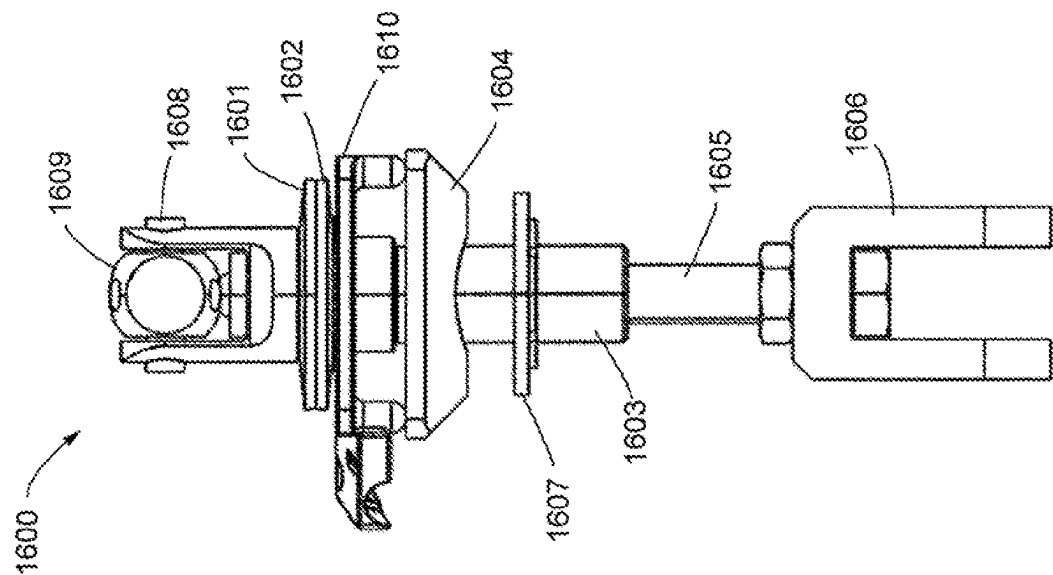


FIG. 22

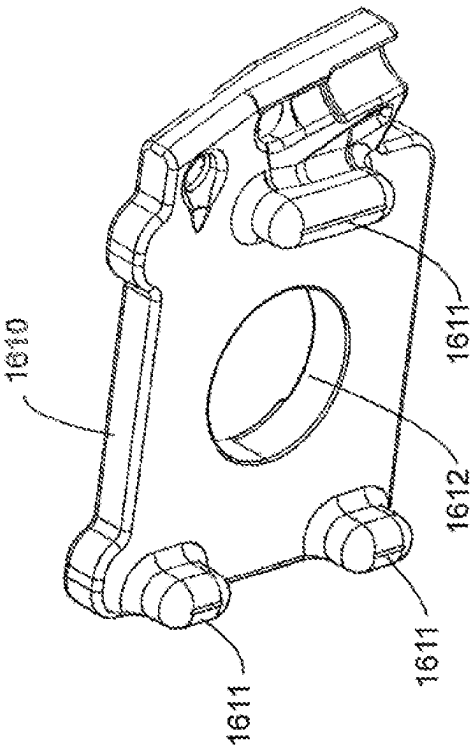


FIG. 26

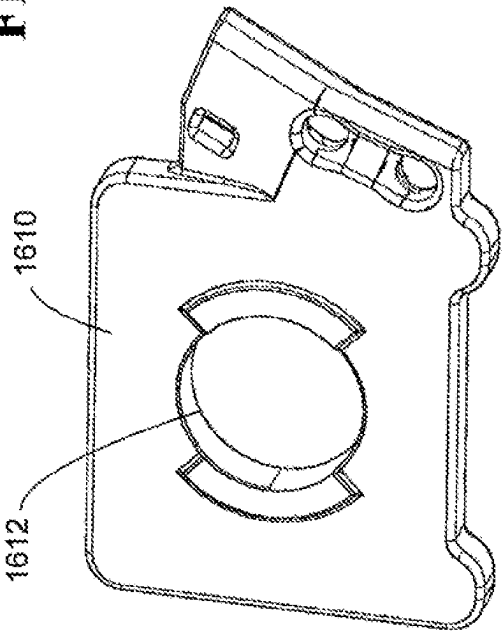


FIG. 25

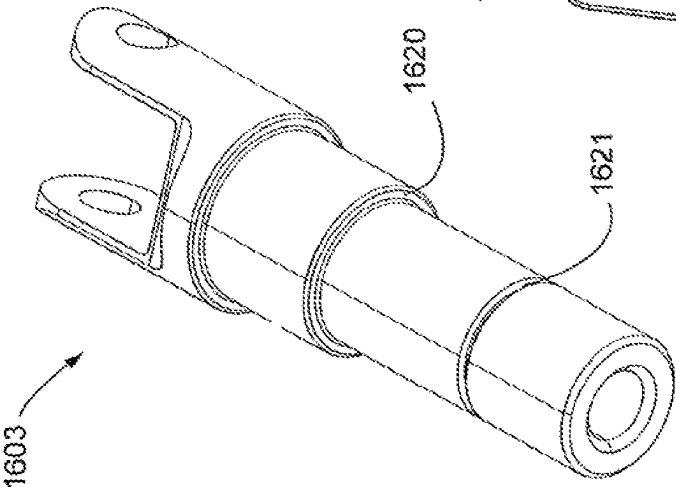


FIG. 24

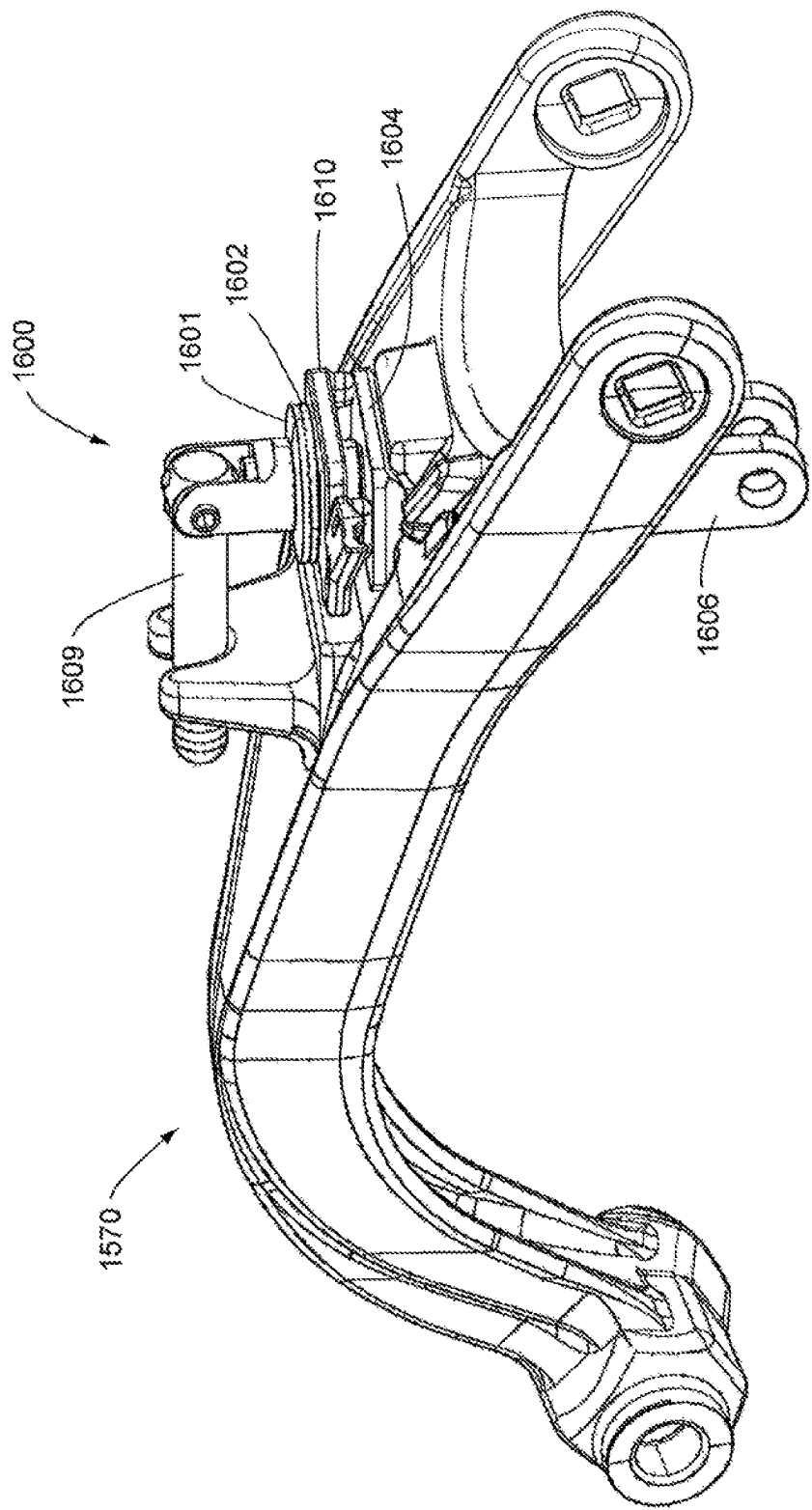
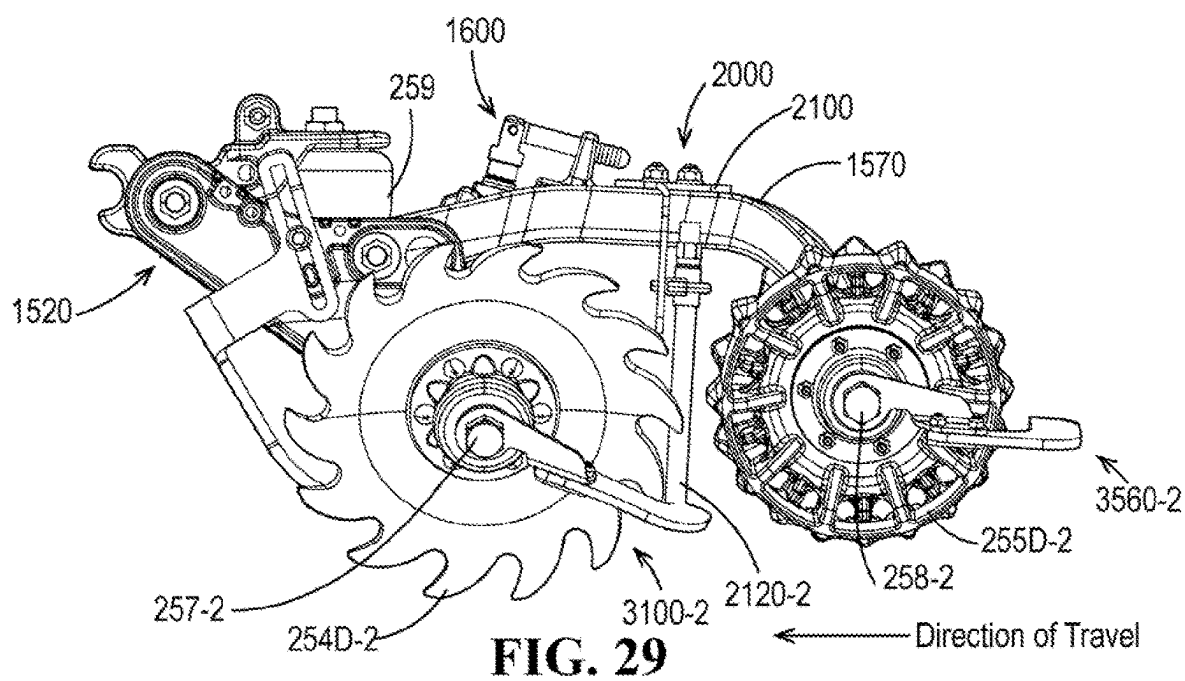
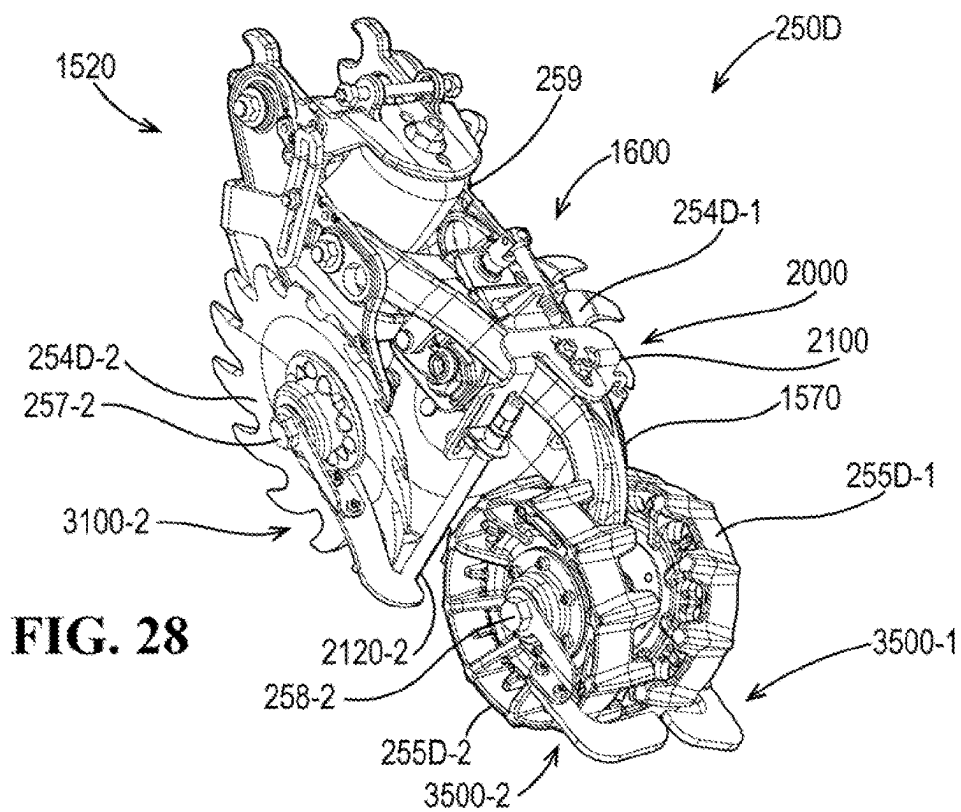


FIG. 27



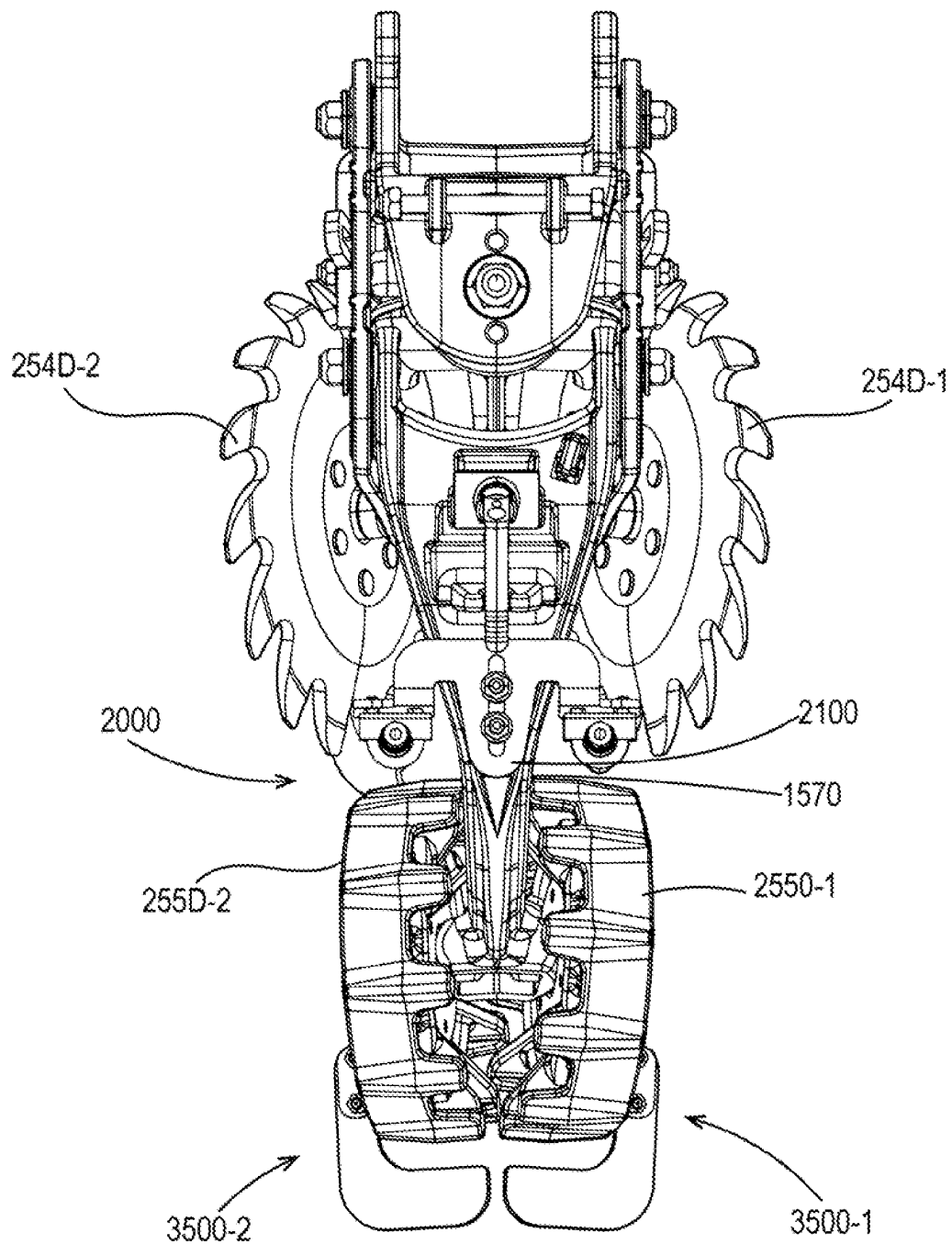


FIG. 30

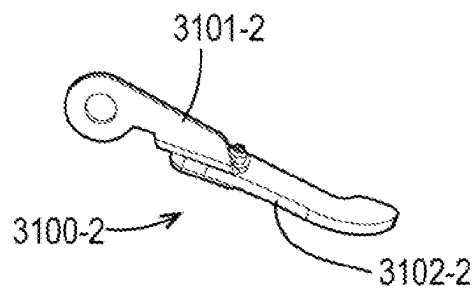


FIG. 31

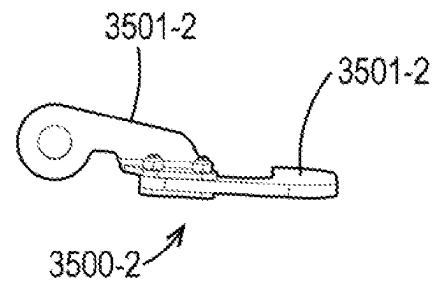


FIG. 32

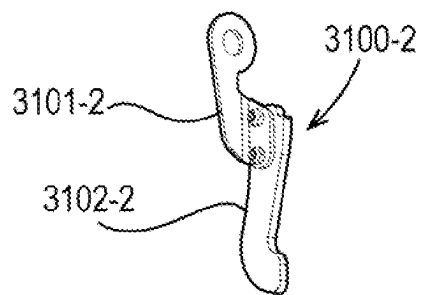


FIG. 33

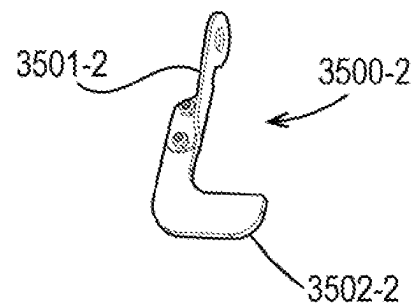


FIG. 34

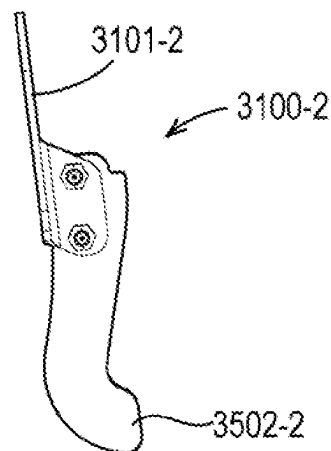


FIG. 35

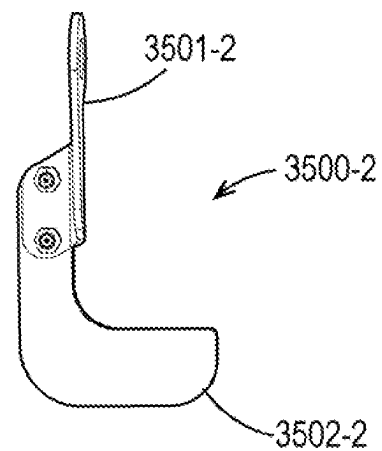


FIG. 36

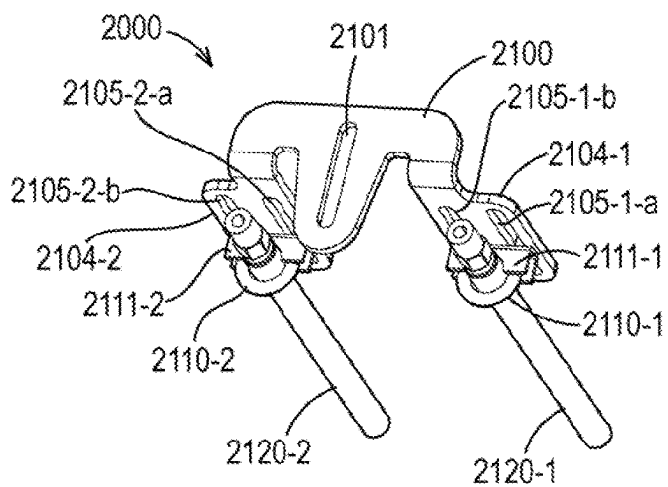


FIG. 37

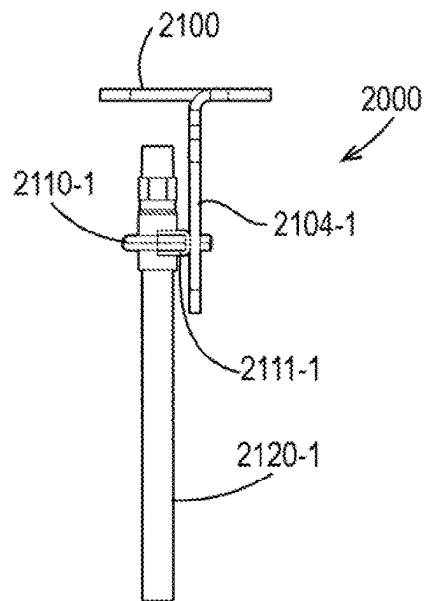


FIG. 38

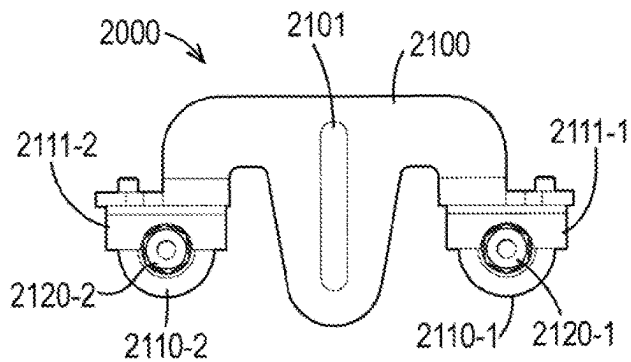


FIG. 39

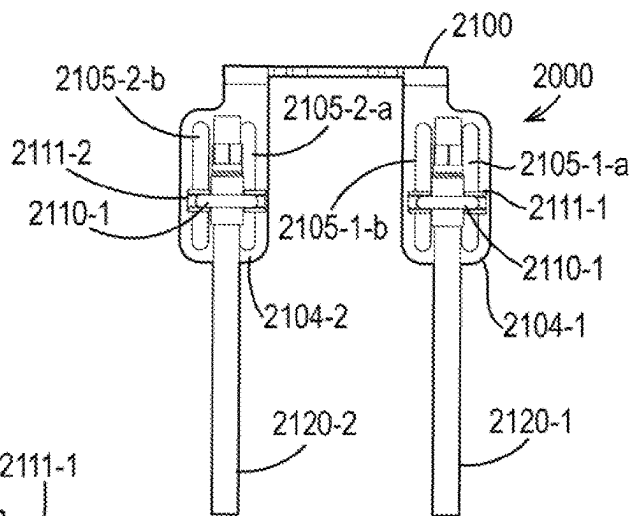


FIG. 40

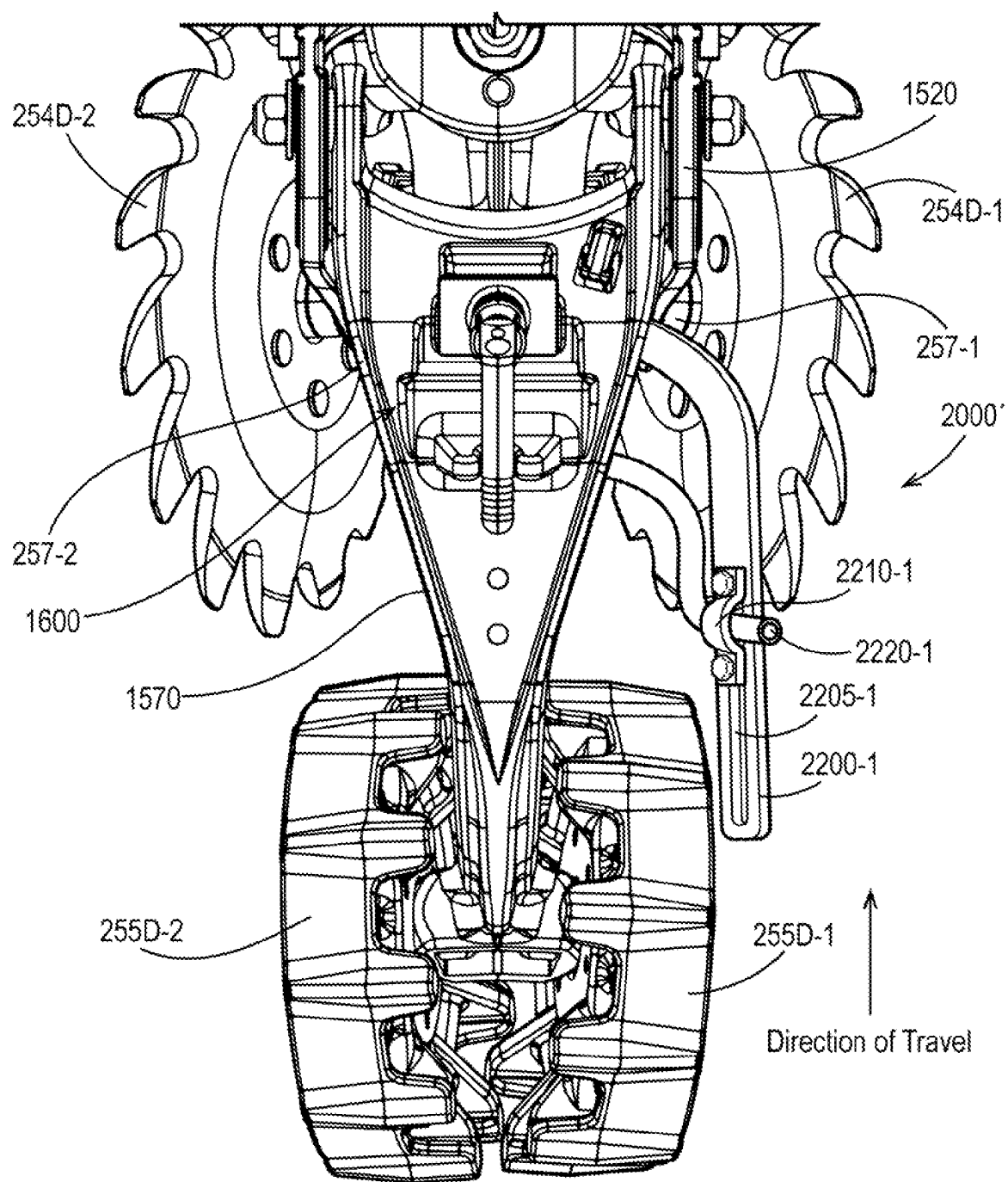


FIG. 41

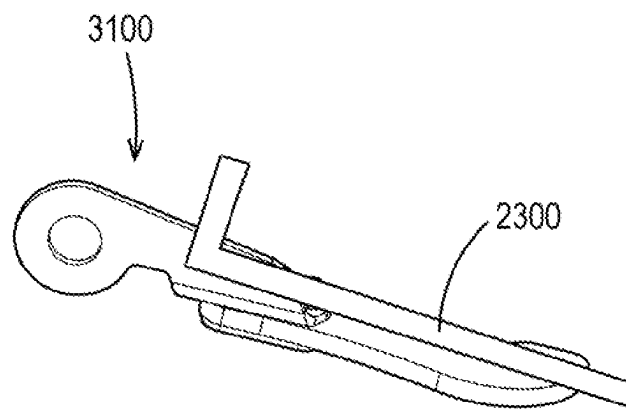


FIG. 42

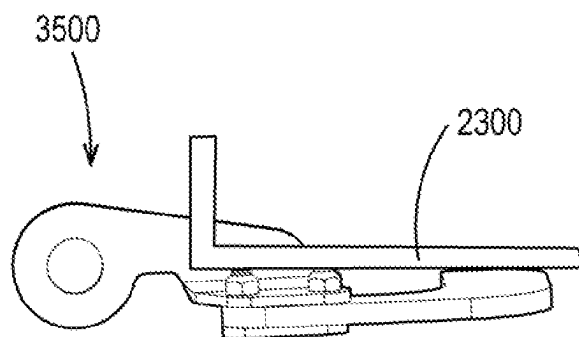


FIG. 43

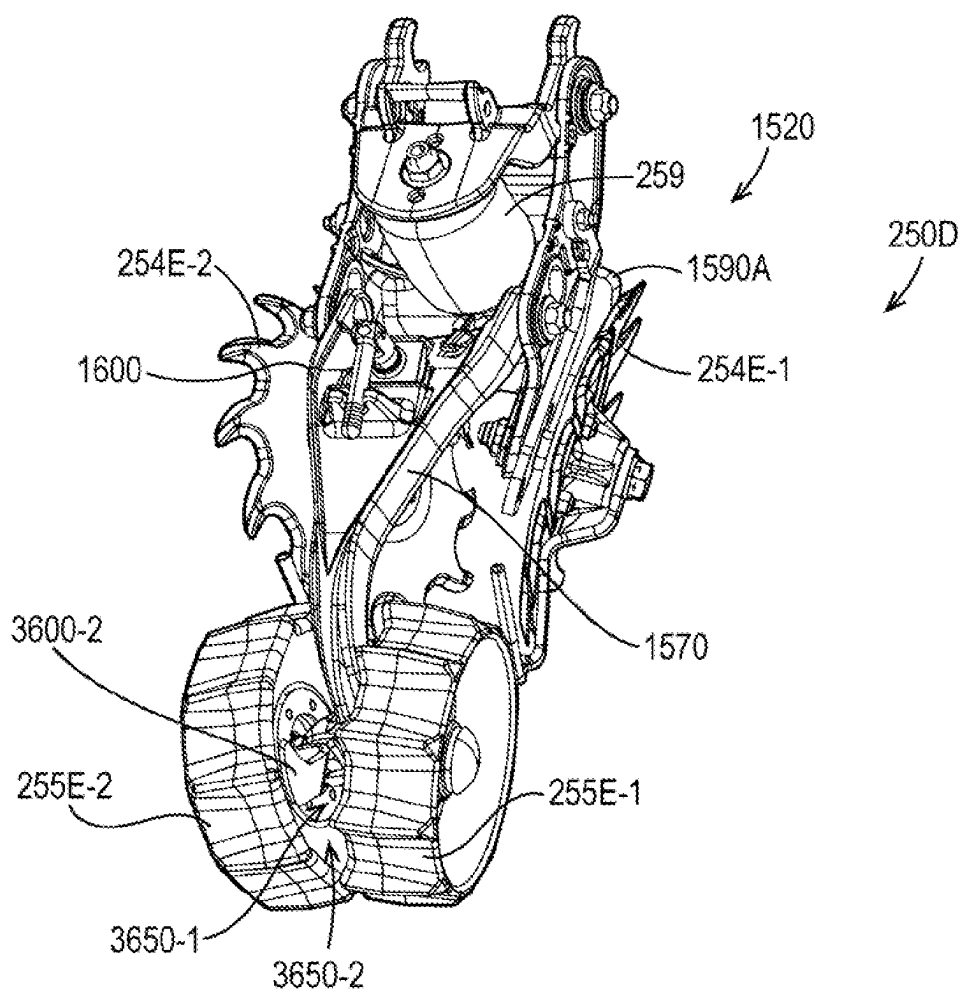


FIG. 44

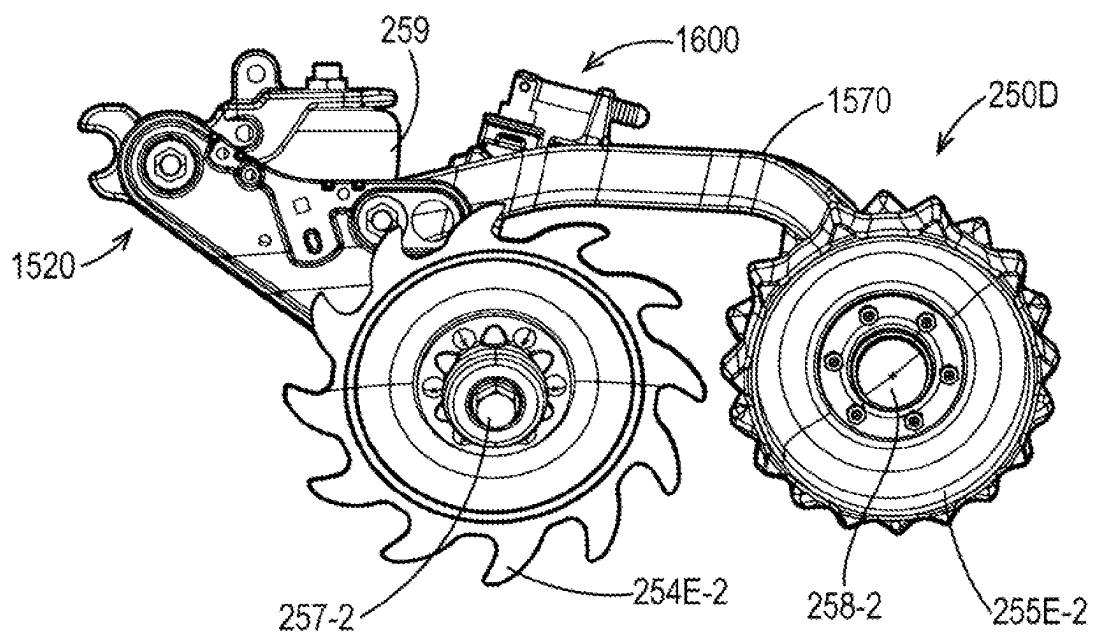
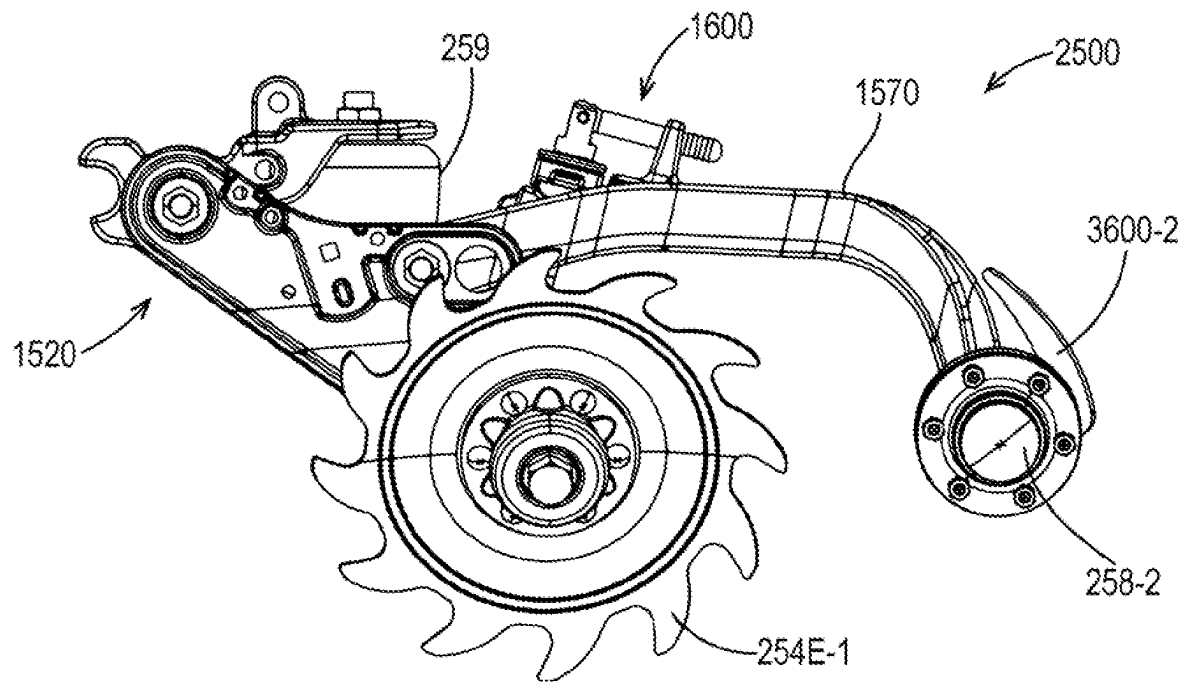
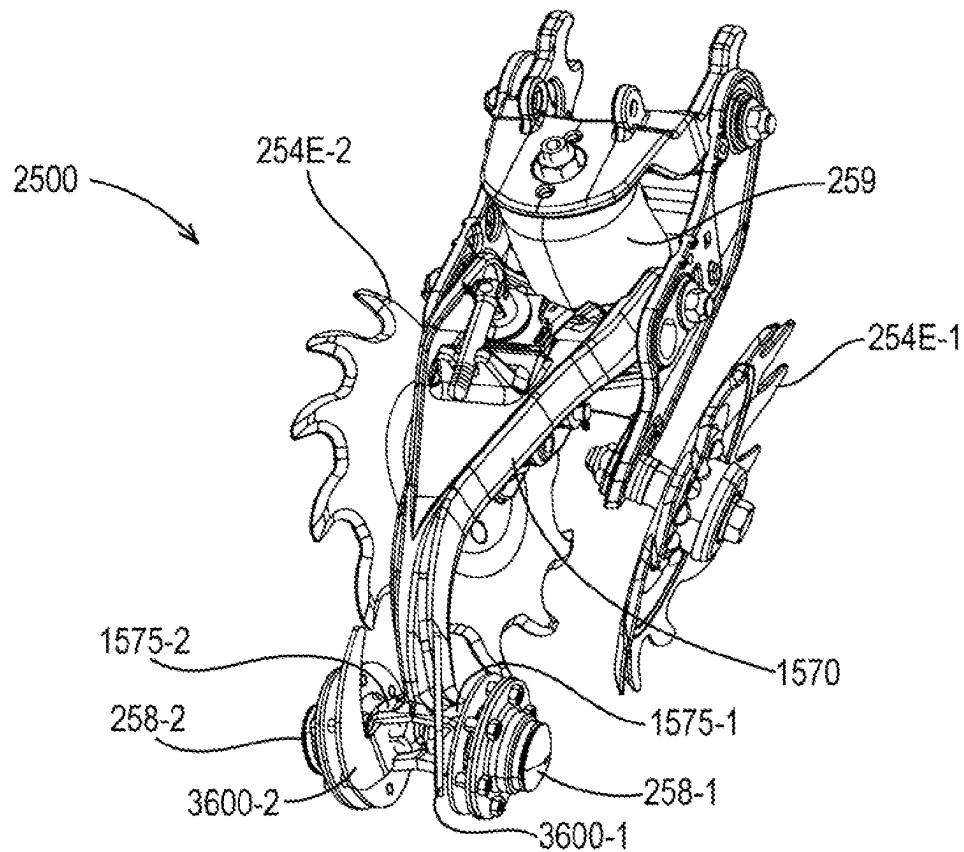


FIG. 45

**FIG. 46****FIG. 47**

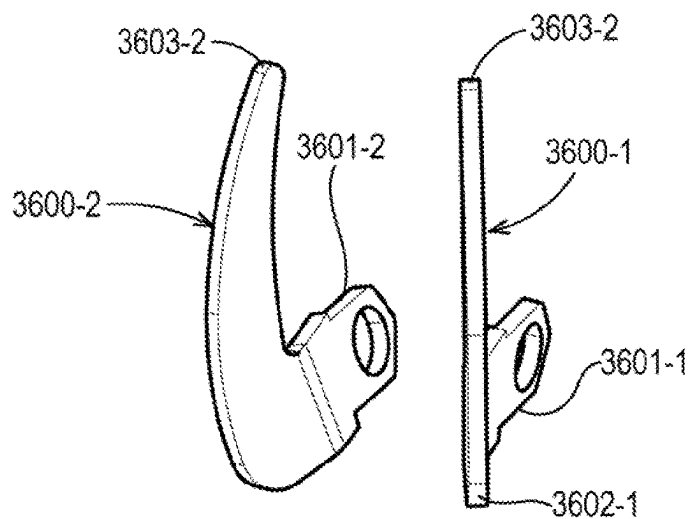


FIG. 48

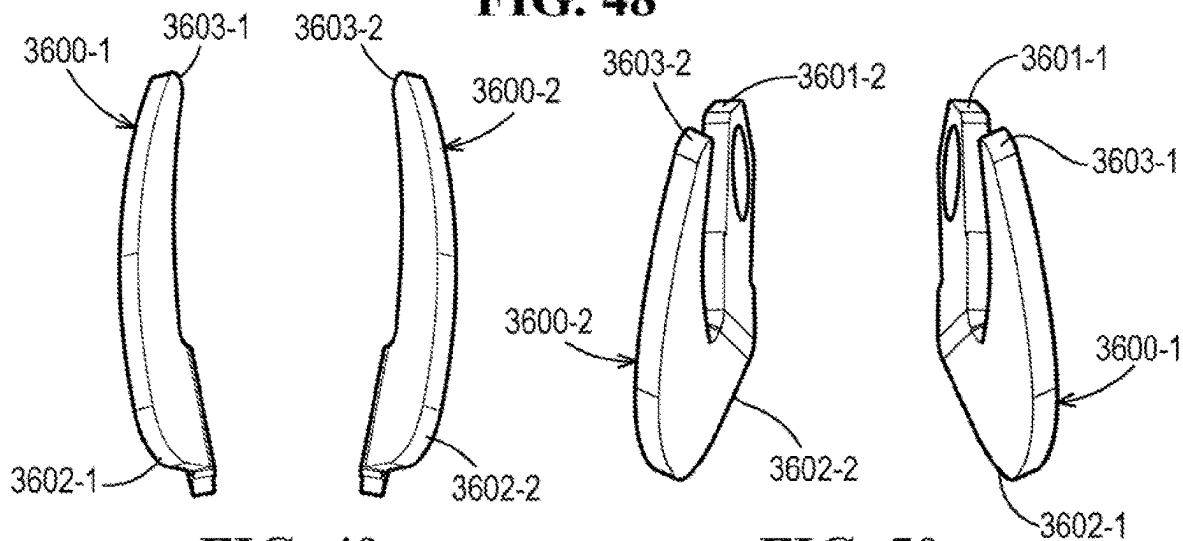


FIG. 49

FIG. 50

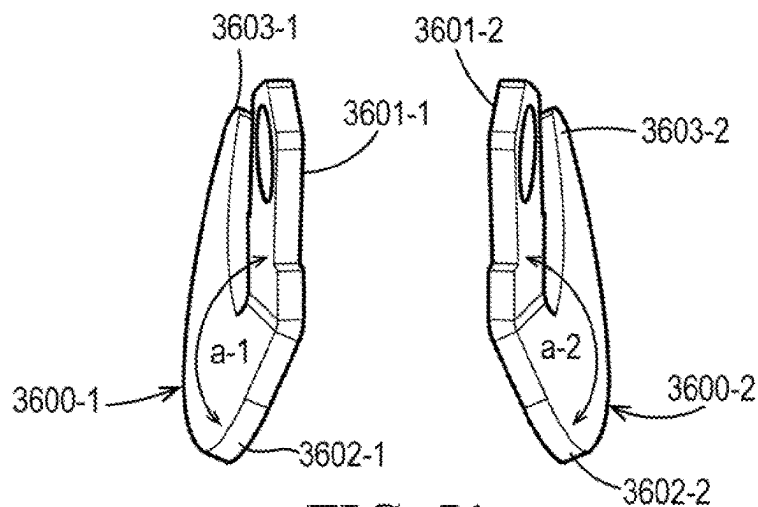


FIG. 51

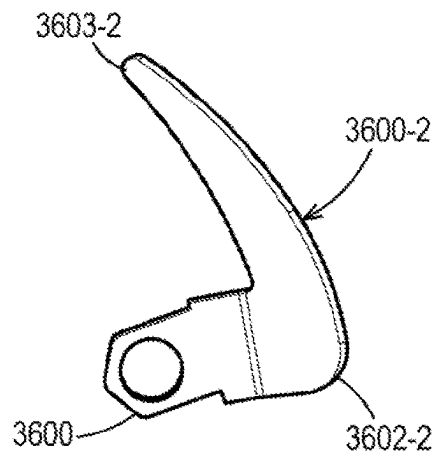


FIG. 52

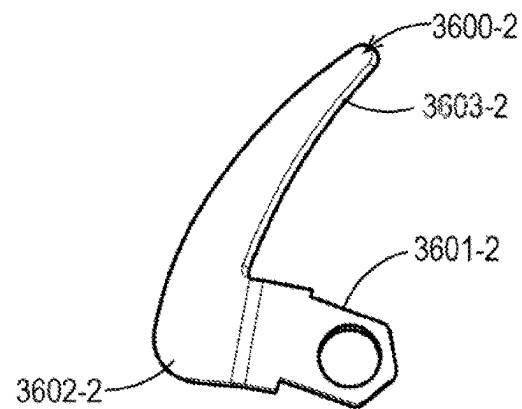


FIG. 53

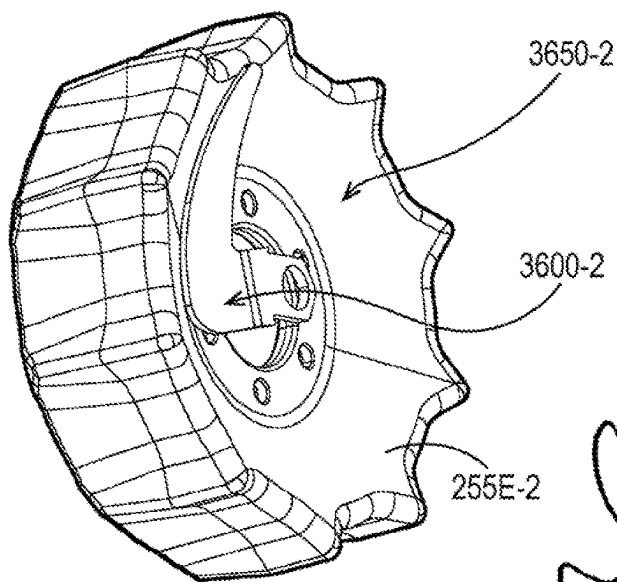


FIG. 54

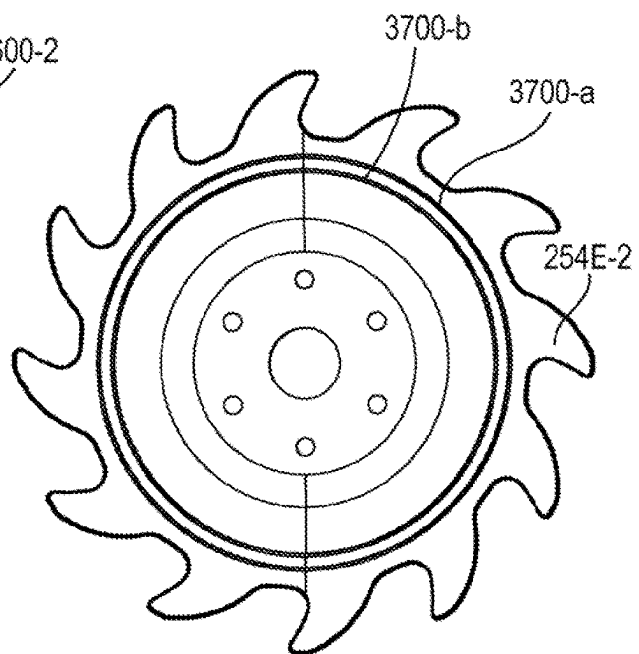


FIG. 55

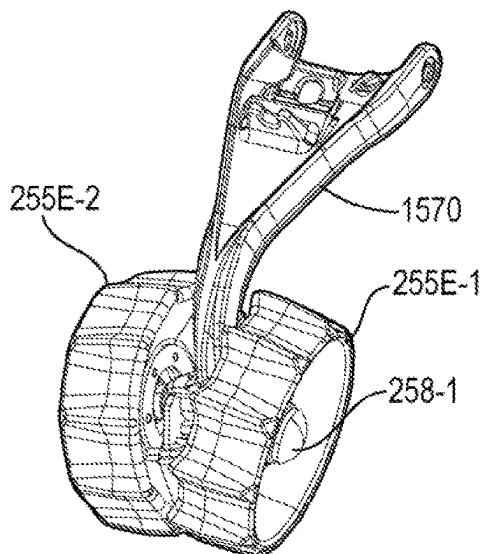


FIG. 56

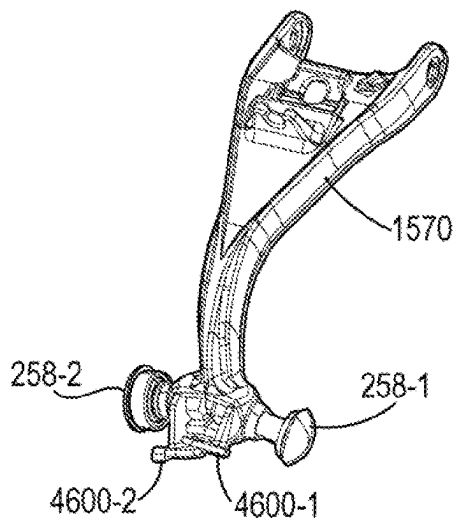


FIG. 57

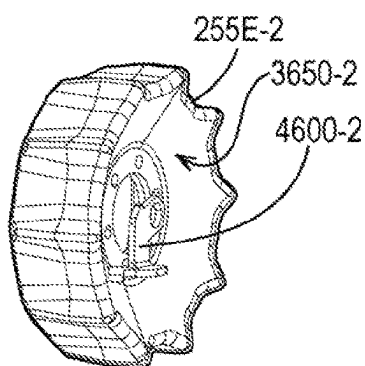


FIG. 58

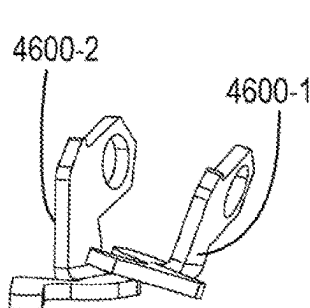


FIG. 59

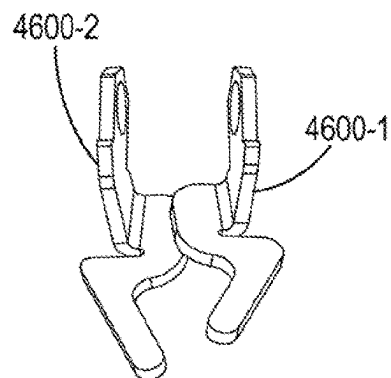


FIG. 60

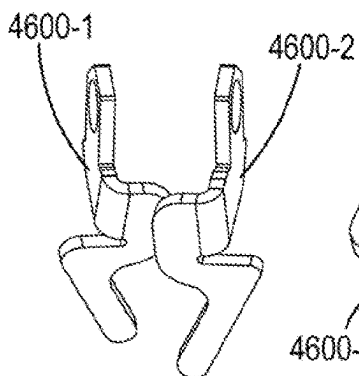


FIG. 61

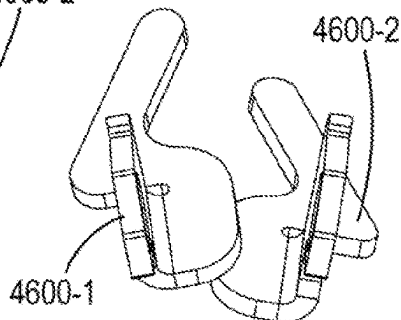


FIG. 62

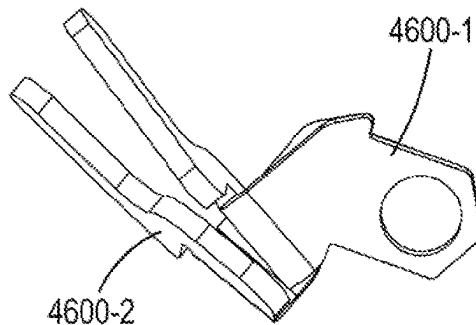


FIG. 63

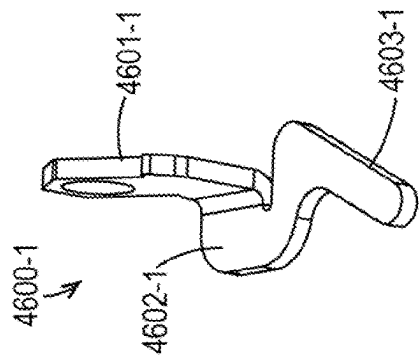


FIG. 64

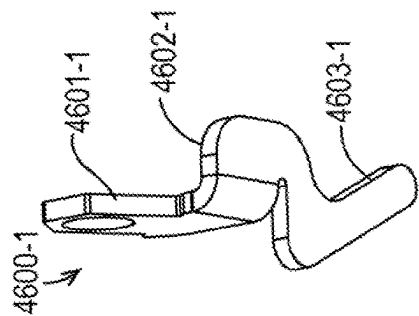


FIG. 65

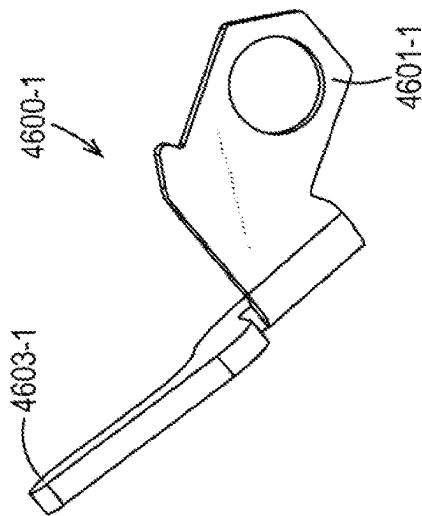


FIG. 66

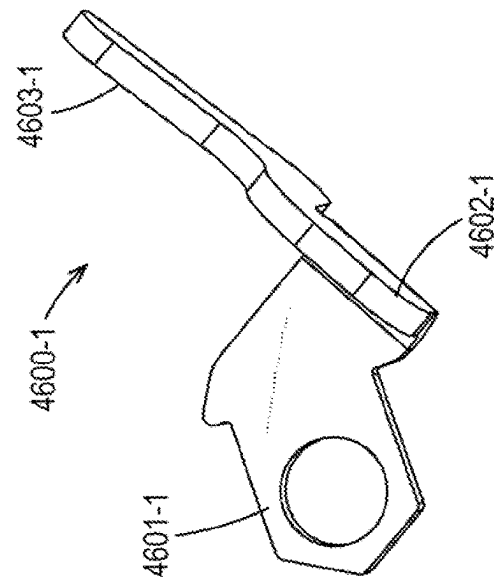


FIG. 67

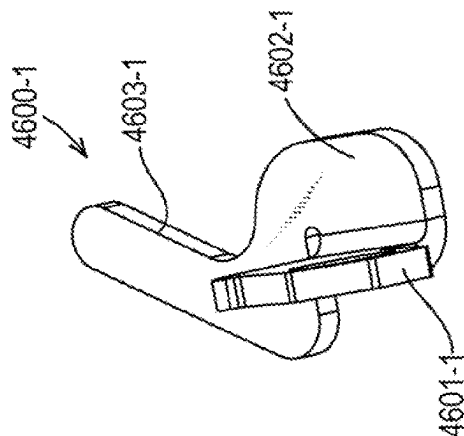


FIG. 68

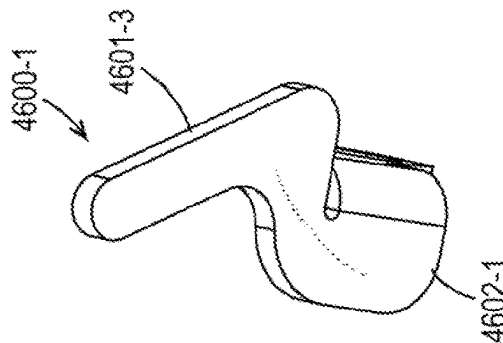


FIG. 69

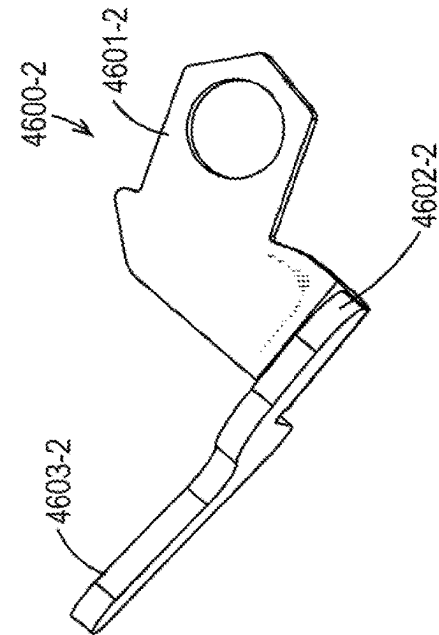


FIG. 70

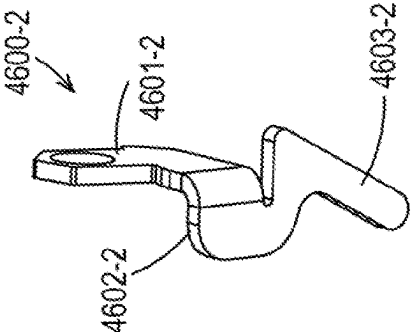


FIG. 71

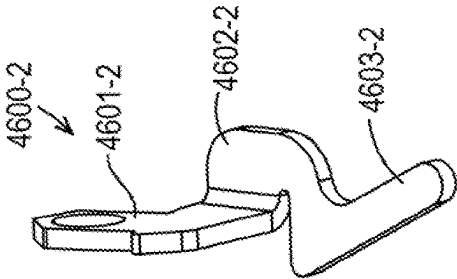


FIG. 72

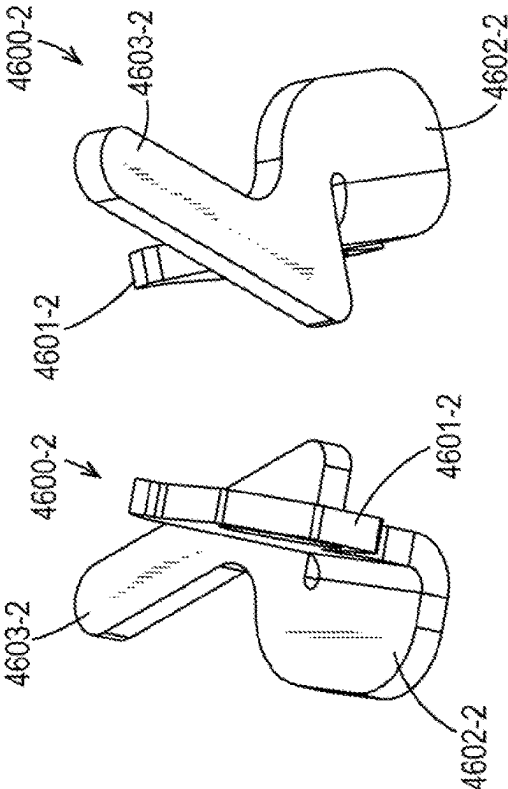


FIG. 73

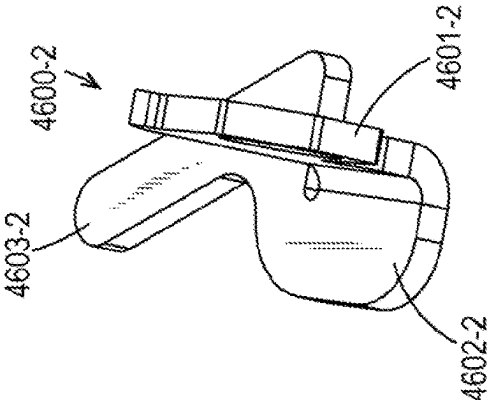


FIG. 74

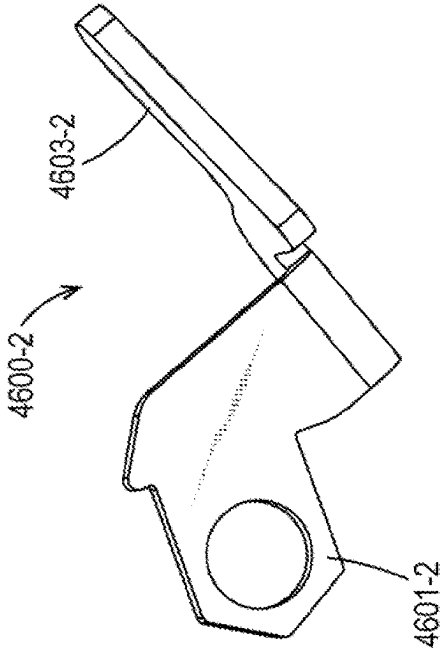


FIG. 75

1

SCRAPER FOR TRENCH CLOSING ASSEMBLY WHEEL

BACKGROUND

It is well known that good seed-to-soil contact within the seed trench is a critical factor in uniform seed emergence and high yields. To obtain good seed-to-soil contact, once seeds are planted in a seed trench, the seed trench needs to be closed so that soil surrounds the seed.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a side elevation view of an embodiment of a row unit of an agricultural planter.

FIG. 2 is a diagram of a system for implementing operational control of the closing wheel assembly based on signals generated by the trench closing sensor.

FIG. 3 is a left side perspective view of an embodiment of a trench closing assembly.

FIG. 4 is a top plan view of the trench closing assembly of FIG. 3.

FIG. 5 is a bottom plan view of the trench closing assembly of FIG. 3.

FIG. 6 is a top perspective view of the trench closing assembly of FIG. 3 with some parts removed.

FIG. 7 is a bottom perspective view of the trench closing assembly of FIG. 6.

FIG. 8 is a bottom perspective view of the trench closing assembly of FIG. 7 with some parts removed.

FIG. 9 is a bottom perspective view of the trench closing assembly of FIG. 8 with some parts removed.

FIG. 10 is a top perspective view of the trench closing assembly of FIG. 9.

FIG. 11 is a top perspective view of the frame of the trench closing assembly of FIG. 3.

FIG. 12 is a bottom perspective view of the frame of FIG. 11.

FIG. 13 is bottom plan view of the frame of FIG. 11.

FIG. 14 is top perspective view of the frame of the trench closing assembly of FIG. 3.

FIG. 15 is a bottom perspective view of the frame of FIG. 14.

FIG. 16 is a top perspective view of the connection bracket of the frame of FIG. 14.

FIG. 17 is a bottom perspective view of the connection bracket of FIG. 16.

FIG. 18 is a top perspective view of the attachment bracket of the frame of FIG. 14.

FIG. 19 is a bottom perspective view of the attachment bracket of FIG. 15.

FIG. 20 is top perspective view of the mounting arm of the trench closing assembly of FIG. 3.

FIG. 21 is a bottom perspective view of the mounting arm of FIG. 20.

FIG. 22 is a front elevation view of the handle assembly of the trench closing assembly of FIG. 3.

FIG. 23 is a left side elevation view of the handle assembly of FIG. 22.

FIG. 24 is a perspective view of the shank of the handle assembly of FIG. 22.

FIG. 25 is a bottom perspective view of the pressure sensor of the handle assembly of FIG. 22.

FIG. 26 is a top perspective view of the pressure sensor of FIG. 25.

FIG. 27 is a perspective view of the handle assembly and mounting arm of the trench closing assembly of FIG. 3.

2

FIG. 28 is a perspective view of the trench closing assembly of FIG. 3 further including soil deflectors and a fluid applicator according to an embodiment.

FIG. 29 is a left side elevation view of the trench closing assembly of FIG. 28.

FIG. 30 is a top plan view of the trench closing assembly of FIG. 28.

FIG. 31 is a left side elevation view of the closing wheel soil deflector of FIG. 28.

FIG. 32 is a left side elevation view of the press wheel soil deflector of FIG. 28.

FIG. 33 is a perspective view of the closing wheel soil deflector of FIG. 31.

FIG. 34 is a perspective view of the press wheel soil deflector of FIG. 32.

FIG. 35 is a top plan view of the closing wheel soil deflector of FIG. 31.

FIG. 36 is a top plan view of the press wheel soil deflector of FIG. 32.

FIG. 37 is a perspective view of the fluid applicator according to an embodiment.

FIG. 38 is a right side elevation view of the fluid applicator of FIG. 37.

FIG. 39 is a top plan view of the fluid applicator of FIG. 37.

FIG. 40 is a rear elevation view of the fluid applicator of FIG. 37.

FIG. 41 is a top plan view of a fluid applicator according to an embodiment.

FIG. 42 is a left side elevation view of the closing wheel soil deflector of FIG. 31 and including a nozzle.

FIG. 43 is a left side elevation view of the closing wheel soil deflector of FIG. 32 and including a nozzle.

FIG. 44 is a perspective view of the trench closing assembly of FIG. 3 with modified closing wheels and scrapers for the press wheels according to an embodiment.

FIG. 45 is a left side elevation view of the trench closing assembly of FIG. 44.

FIG. 46 is a left side elevation view of the trench closing assembly of FIG. 45 with the press wheels removed.

FIG. 47 is a perspective view of the trench closing assembly of FIG. 46.

FIG. 48 is a perspective view of the scrapers.

FIG. 49 is a front elevation view of the scrapers of FIG. 48.

FIG. 50 is a top plan view of the scrapers of FIG. 48.

FIG. 51 is a bottom plan view of the scrapers of FIG. 48.

FIG. 52 is a left side elevation view of the left scraper of FIG. 48.

FIG. 53 is a right side elevation view of the left scraper of FIG. 48.

FIG. 54 is a perspective view of the left press wheel and left scraper of FIG. 44.

FIG. 55 is a left side elevation view of the closing wheel of FIG. 44.

FIG. 56 is a partial perspective view of the trench closing assembly of FIG. 3 with modified scrapers for the press wheels according to an embodiment.

FIG. 57 is a perspective view of the trench closing assembly of FIG. 56 with the press wheels removed for viewing.

FIG. 58 is a perspective view of a press wheel and scraper of FIG. 56.

FIG. 59 is a perspective view of the scrapers of FIG. 56.

FIG. 60 is a top plan view of the scrapers of FIG. 59.

FIG. 61 is a bottom plan view of the scrapers of FIG. 59.

3

FIG. 62 is a front elevation view of the scrapers of FIG. 59.

FIG. 63 is a right side elevation view of the scrapers of FIG. 59.

FIG. 64 is top plan view of the right scraper of FIG. 59.

FIG. 65 is a bottom plan view of the right scraper of FIG. 64.

FIG. 66 is a right side elevation view of the right scraper of FIG. 64.

FIG. 67 is a left side elevation view of the right scraper of FIG. 64.

FIG. 68 is a front elevation view of the right scraper of FIG. 64.

FIG. 69 is a rear elevation view of the right scraper of FIG. 64.

FIG. 70 is a top plan view of the left scraper of FIG. 59.

FIG. 71 is a bottom plan view of the left scraper of FIG. 70.

FIG. 72 is a right side elevation view of the left scraper of FIG. 70.

FIG. 73 is a left side elevation view of the left scraper of FIG. 70.

FIG. 74 is a front elevation view of the left scraper of FIG. 70.

FIG. 75 is a rear elevation view of the left scraper of FIG. 70.

DESCRIPTION

All references cited herein are hereby incorporated by reference in their entireties. In the event of a conflict in a definition in the present disclosure and that of a cited reference, the present disclosure controls.

Referring now to the drawings, wherein like reference numerals designate identical or corresponding parts throughout the several views, FIG. 1 illustrates an embodiment of an agricultural planter row unit 200. The row unit 200 is comprised of a frame 204 pivotally connected to a toolbar 202 by a parallel linkage 206 enabling each row unit 200 to move vertically independently of the toolbar 202. The frame 204 operably supports one or more hoppers 208, a seed meter 210, a seed delivery mechanism 212, a downforce control system 214, a seed trench opening assembly 220, a trench closing assembly 250, an optional packer wheel assembly 260, and an optional row cleaner assembly 270. It should be understood that the row unit 200 shown in FIG. 1 may be for a conventional planter or the row unit 200 may be a central fill planter, in which case the hoppers 208 may be replaced with one or more mini-hoppers and the frame 204 modified accordingly as would be recognized by those of skill in the art.

The optional downforce control system 214 is disposed to apply lift and/or downforce on the row unit 200 such as disclosed in U.S. Publication No. US2014/0090585. The downforce applied by downforce control system 214 can be determined by methods disclosed in U.S. Pat. No. 9,173,339.

The seed trench opening assembly 220 includes a pair of opening discs 222 rotatably supported by a downwardly extending shank member 205 of the frame 204. The opening discs 222 are arranged to diverge outwardly and rearwardly so as to open a v-shaped trench 10 in the soil 11 as the planter traverses the field. The seed delivery mechanism 212, such as a seed tube or seed conveyor, is positioned between the opening discs 222 to deliver seed from the seed meter 210 into the opened seed trench 10. The depth of the seed trench 10 is controlled by a pair of gauge wheels 224

4

positioned adjacent to the opening discs 222. The gauge wheels 224 are rotatably supported by gauge wheel arms 226 which are pivotally secured at one end to the frame 204 about pivot pin 228. A rocker arm 230 is pivotally supported on the frame 204 by a pivot pin 232. It should be appreciated that rotation of the rocker arm 230 about the pivot pin 232 sets the depth of the trench 10 by limiting the upward travel of the gauge wheel arms 226 (and thus the gauge wheels) relative to the opening discs 222. The rocker arm 230 may be adjustably positioned via a linear actuator 234 mounted to the row unit frame 204 and pivotally coupled to an upper end of the rocker arm 230. The linear actuator 234 may be controlled remotely or automatically actuated as disclosed, for example, in International Publication No. WO2014/186810.

An optional downforce sensor 238 is configured to generate a signal related to the amount of force imposed by the gauge wheels 224 on the soil. In some embodiments the pivot pin 232 for the rocker arm 230 may comprise the downforce sensor 238, such as the instrumented pins disclosed in U.S. Pat. No. 8,561,472.

An optional seed meter 210 may be any commercially available seed meter, such as a finger-type meter or a vacuum seed meter. One exemplary vacuum seed meter is the VSet® meter, available from Precision Planting LLC, 23207 Townline Rd, Tremont, IL 61568.

An optional packer wheel assembly 260 comprises an arm 262 pivotally attached to the row unit frame 204 and extends rearward of the closing wheel assembly 250 and in alignment therewith. The arm 262 rotatably supports a packer wheel 264. An actuator 266 is pivotally attached at one end to the arm 262 and at its other end to the row unit frame 204 to vary the amount of downforce exerted by the packer wheel 264 to pack the soil over the seed trench 10.

An optional row cleaner assembly 270 may be the Clean-Sweep® system available from Precision Planting LLC, 23207 Townline Rd, Tremont, IL 61568. The row cleaner assembly 270 includes an arm 272 pivotally attached to the forward end of the row unit frame 204 and aligned with the trench opening assembly 220. A pair of row cleaner wheels 274 are rotatably attached to the forward end of the arm 272. An actuator 276 is pivotally attached at one end to the arm 272 and at its other end to the row unit frame 204 to adjust the downforce on the arm to vary the aggressiveness of the action of the row cleaning wheels 274 depending on the amount of crop residue and soil conditions.

Referring to FIG. 2, a monitor 300 is visible to an operator within the cab of a tractor pulling the planter. The monitor 300 may be in signal communication with a GPS unit 310, the trench closing assembly actuator 256 and the optional packer wheel assembly actuator 266 to enable operational control of the trench closing assembly 250 and the optional packer wheel assembly 260 based on the signals generated by trench closing sensors 1000, which are described in International Publication No. WO2017/197274. Also as discussed later, the monitor 300 may be programmed to display operational recommendations based on the signals generated by the trench closing sensors 1000. The monitor 300 may also be in signal communication with the row cleaner actuator 276, the downforce control system 214, the depth adjustment actuator 234 to enable operational control of row cleaner assembly 270, the downforce control system 214 and the trench opening assembly 230, respectively.

Trench Closing Assembly

Examples of Trench Closing Assemblies can be found in U.S. Patent Application Nos. 62/637,372, filed 1 Mar. 2018, 62/644,201, filed 16 Mar. 2018, 62/731,813, filed 14 Sep.

2018, 62/791,203, filed on 11 Jan. 2019, 62/831,338, filed 9 Apr. 2019, 62/843,037, filed 3 May 2019, and in International Application No. PCT/US2019/020452, filed on 2 Mar. 2019. Also, an example of a trench closing assembly is the FurrowForce™ trench closing assembly available from Precision Planting, LLC.

FIGS. 3-27 illustrate a trench closing assembly 250D according to an embodiment. In this embodiment, trench closing assembly 250D has a main frame 251D that is connected to row unit 200. Trench closing assembly 250D has a frame 1520, an actuator 259, a pair of closing wheels 254D-1, 254D-2, and optionally, a press wheel 255D.

Turning to FIGS. 14-19, frame 251D has a connection bracket 1540 and an attachment bracket 1550. Connection bracket 1540 can have one or more bolts 1209D extending through apertures 1207D in connection bracket 1540 for mounting the frame 251D to the row unit 200. Connection bracket 1540 (as with connection brackets 1200 and 1201A) can be varied to mate attachment to different styles of row units. Connection bracket 1540 has a first side 1542-1, a second side 1542-2, and a plate 1543 disposed between first side 1542-1 and second side 1542-2. First side 1542-1 and second side 1542-2 each have a post 1541-1 and 1541-2, respectively, protruding perpendicularly outward.

Attachment bracket 1550 connects to connection bracket 1540. Attachment bracket 1550 provides a common structure for mounting other parts while connection bracket 1540 has a varied structure to mate with different styles of row units. Attachment bracket has a first side 1552-1, a second side 1552-2, a crossbar 1557 disposed between first side 1552-1 and second side 1552-2, and plate 1553 disposed between first side 1552-1 and second side 1552-2, and plate 1553. First side 1552-1 and second side 1552-2 each have a u-shaped opening 1555-1 and 1555-2, respectively, for connection to posts 1541-1 and 1541-2, respectively. First side 1552-1 and second side 1552-2 have openings 1558-1 and 1558-2, respectively, for accepting pivots 1529-1 and 1529-2, respectively. Attachment bracket 1550 can be secured to connection bracket with fastener 1549. Optionally, attachment bracket 1550 can also have an opening 1501 disposed in first side 1552-1 or second side 1552-2 for accepting a pin. While connection bracket 1540 is illustrated with separate parts it may be fabricated as a unitary part.

Optionally, as illustrated in FIGS. 5 and 7, a guard 1590 can be disposed on frame 1520 ahead of closing wheels 254D-1, 254D-2. Guard 1590 can prevent rocks, rootballs, or other trash from approaching closing wheels 254D-1, 254D-2. The height of guard 1590 can be adjusted by changing the placement of arms 1591-1 and 1591-2 on frame 1520.

Turning to FIGS. 20-21, mounting arm 1570 has a first side 1571-1, a second side 1571-2, a plate 1574 disposed between first side 1571-1 and second side 1571-2, a hole 1573 through plate 1574 for passage of sleeve 1603, axle hubs 1575-1 and 1575-2 for mounting press wheels 255D-1 and 255D-2, respectively, and, optionally, a handle cradle 1579.

Turning to FIGS. 6-13, frame 1520 is pivotally connected to frame 251D through pivots 1529-1 and 1529-2. Frame 1520 has a first side 1526-1 and a second side 1526-2, a plate 1525 connecting first side 1526-1 and second side 1526-2, a connection arm 1527 extending rearwardly along the direction of travel from plate 1525, a cross-brace 1528-1 connecting connection arm 1527 to first arm 1526-1, a cross-brace 1528-2 connecting connection arm 1527 to second arm 1526-2. First side 1526-1 and second side 1526-2 have openings 1521-1 and 1521-2 for disposing about pivots

1529-1 and 1529-2, respectively. Mounting arm 1570 can attach to frame 1520 at connections 1522-1a and 1522-2a or to 1522-1b and 1522-2b. The plurality of connections allows the distance between closing wheels 254D-1, 254D-2, and press wheel 255D (255D-1, 255D-2) to be changed. There can be one connection 1522 or a plurality of connections 1522. As the mounting arm 1570 position is changed, there are also corresponding connections 1523-a and 1523-b (matching in number to connection 1522) for connecting handle assembly 1600 to frame 1520. Frame 1520 also has connections 1524 (1524-1a, 1524-1b, 1524-2a, and 1524-2b) for mounting closing wheels 254D-1, 254D-2. While there can be one set of connections 1524, the plurality of connections 1524 allow for forward and back placement of the closing wheels 254D-1, 254D-2 on frame 1520, or closing wheels 254D-1, 254D-2 can be offset from each other with one closing wheels 254D-1, 254D-2 being mounted to a forward location (the “a” position) or to a rear location (the “b” position). As illustrated, closing wheels 254D-1, 254D-2 are offset from each other. Optionally, frame 1520 can have an opening 1502 in either the first side 1526-1 or second side 1526-2 (shown in 1526-2) for accepting a pin. Frame 1520 can be raised to allow opening 1501 and opening 1502 to align for accepting a pin (not shown). This allows trench closing assembly 250D to be raised for transport or when closing is not needed. While illustrated on one side, openings 1501 and 1502 can be disposed on both sides.

Actuator 259 is disposed between plate 1525 and plate 1553 to apply a force to plate 1553 to cause frame 1520 to pivot and apply pressure to closing wheels 254D-1 and 254D-2.

Turning to FIGS. 22-27, handle assembly 1600 is illustrated. Handle assembly 1600 has a sleeve 1603 having a first diameter 1620 and a second diameter 1621. Second diameter 1621 is small enough to be disposed through load sensor 1610, and first diameter 1620 is large enough so that it cannot pass through load sensor 1610. Sleeve 1603 has a bracket 1622 (u-shaped bracket as illustrated or any other shape) for accepting handle 1609. Disposed on sleeve 1603 below bracket 1622 are bevel washers 1601 and 1602. Bevel washer 1601 and bevel washer 1602 are disposed with their concave surfaces facing each other. This allows bevel washers 1601 and 1602 to flex to absorb shocks experienced by trench closing assembly 250D to prevent overloading of load sensor 1610. Before bevel washers 1601 and 1602 reach maximum flex, first diameter 1620 will contact plate 1604 to limit the travel. Load sensor 1610 is a pancake load sensor. Load sensor 1610 has a hole 1612 for passage of sleeve 1603. Disposed on the underside of load sensor 1610 are a plurality of feet 1611 to allow load sensor 1610 to flex and measure force. Load sensor 1610 can be disposed on mounting arm 1570 directly, or as shown, a plate 1604 can be disposed between load sensor 1610 and mounting arm 1570. Optionally, a washer 1607 can be disposed about sleeve 1603 under mounting arm 1570. Handle assembly 1600 is connected to frame 1520 at connection 1523-a or 1523-b with a bracket 1606 (u-shaped bracket) and a bolt 1605 connecting bracket 1606 with sleeve 1603. In the horizontal position, handle 1609 locks handle assembly 1600 in place against mounting arm 1570. In the vertical position, handle 1609 releases handle assembly from engagement with mounting arm 1570. Bolt 1605 can be adjusted to set a vertical placement of mounting arm 1570 relative to frame 1520.

Load sensor 1610 can be connected to a network directly through a plug (not shown) having a CAN processor to allow direct communication over a CAN network. The CAN

processor can communicate pressure readings and provide control signals over the CAN network. Alternatively, load sensor **1610** can be connected to a control module (either an on-row module, or a module controlling a plurality of row) to communicate pressure readings that are then processed by the control module.

Soil Deflector

FIGS. **28-30** illustrate soil deflectors (**3100**, **3500**) and fluid applicator (**2000**) disposed on closing assembly **250D**. Soil deflector **3100** is illustrated on the left, and the right (not shown) is the mirror image of the left. Soil deflectors **3100** and soil deflectors **3500** are illustrated as two parts, but they can be made unitary parts. Soil deflectors **3100** and **3500** can keep dirt/mud from spraying or “rooster tailing” as the trench closing system **250D** traverses the field. Also, soil deflectors **3100** and **3500** can scrape closing wheels **254D** and/or press wheels **255D**, respectively, by being adjusted to contact closing wheels **254D** and/or press wheels **255D**.

Soil deflector **3100-2** is connected to axle **257-2** and extends rearward opposite the direction of travel adjacent to closing wheel **254D-2**. Depending on the shape of closing wheel **254D**, soil deflector **3100** can be shaped to follow the contour of closing wheel **254D** or contact closing wheel **254D**. Soil deflector **3100-2** can be made of two parts, bracket **3101-2** and deflector **3102-2**. When connected to axle **257-2**, optionally a spacer (not shown), such as washer or bushing, can be disposed on axle **257-2** to vary the spacing between soil deflector **3100-2** and closing wheel **254D-2**. Optionally, a connection orientation between bracket **3102-2** and deflector **3102-2** can be changed to change the orientation of deflector **3102-2** with closing wheel **254D-2**. In one embodiment, soil deflector **3100** extends beyond the circumference of closing wheel **254D**, and optionally, extends behind closing wheel **254D** toward frame **1520**. Deflector **3102-2** can be made from a flexible material. A flexible material is a material that will deflect when impacted by soil when trench closing assembly **250D** is operated at typical planting speeds (e.g., up to 20 mph (32 kph) or up to 10 mph (16 kph)). In one embodiment, deflector **3102-2** is polyurethane.

Optionally, soil deflector **3500** (**3500-1**, **3500-2**) can be included for press wheels **255D** (**255D-1**, **255D-2**). While illustrated with two press wheels **255D** (**255D-1**, **255D-2**), there can be just one press wheel **255D**. Soil deflector **3500** (**3500-1**, **3500-2**) is connected to axle **258** (**258-1**, **258-2**) and extends rearward opposite the direction of travel adjacent to press wheel **255D** (**255D-1**, **255D-2**). Depending on the shape of press wheel **255D**, soil deflector **3500** can be shaped to follow the contour of press wheel **255D** or press wheel **255D**. Soil deflector **3500-2** can be made of two parts, bracket **3501-2** and deflector **3502-2**. When connected to axle **258-2**, optionally a spacer (not shown), such as washer or bushing, can be disposed on axle **258-2** to vary the spacing between soil deflector **3500-2** and closing wheel **254D-2**. Optionally, a connection orientation between bracket **3502-2** and deflector **3502-2** can be changed to change the orientation of deflector **3502-2** with press wheel **255D-2**. In one embodiment, soil deflector **3500** extend beyond the circumference of press wheel **255D**, and optionally, extend behind press wheel **255D**. Deflector **3502-2** can be made from a flexible material as defined above.

Fluid Applicator

Fluid applicator **2000** is illustrated in FIGS. **37-40** according to one embodiment. Fluid applicator **2000** has a frame **2100** for connection to mounting arm **1570**. An optional slot **2101** can be disposed in frame **2100** to allow for adjustment front to back along the direction of travel. While illustrated

with two arms **2104** (**2104-1**, **2104-2**), fluid applicator **2000** can have just one arm **2014**. Arms **2104** (**2104-1**, **2104-2**) are disposed on either side of mounting arm **1570**. Nozzles **2120** (**2120-1**, **2120-2**) are attached to arms **2104** (**2104-1**, **2104-2**), respectively. Optionally, arms **2104** (**2104-1**, **2104-2**) can have slots **2105** (**2105-1-a**, **2105-1-b**, **2105-2-a**, **2105-2-b**) to allow nozzles **2120** (**2120-1**, **2120-2**) to be adjusted vertically. For adjustable connection, U-bolts **2110** (**2110-1**, **2110-2**) can connect nozzle **2120** (**2120-1**, **2120-2**) to arms **2104** (**2104-1**, **2104-2**) through slots **2105** (**2105-1-a**, **2105-1-b**, **2105-2-a**, **2105-2-b**). Optionally, slots **2105** (**2105-1-a**, **2105-1-b**, **2105-2-a**, **2105-2-b**) can be sized larger than U-bolts **2110** (**2110-1**, **2110-2**) to allow nozzles **2120** (**2120-1**, **2120-2**) to rotate in a direction transverse to the direction of travel so that nozzles **2120** (**2120-1**, **2120-2**) can deposit fluid closer to or further away from the trench. Optionally, frames **2111** (**2111-1**, **2111-2**) can be included to ensure fit of nozzles **2120** (**2120-1**, **2120-2**) to arms **2104** (**2104-1**, **2104-2**), for example when nozzles **2120** (**2120-1**, **2120-2**) are round. Nozzles **2120** (**2120-1**, **2120-2**) are connected to a fluid source (not shown) for applying fluid to the soil surface. Optionally, frame **2100** can be disposed on mounting arm **1570** rotated 180° from the orientation illustrated.

In an alternative embodiment, fluid applicator **2000'** is illustrated in FIG. **41**. While illustrated on the right side, fluid applicator **2000'** can also be installed as a mirror image on the left side. An arm **2200-1** is connected to an axle **257-1** between frame **1520** and closing wheel **254D-1** and extends in a rearward direction away from the direction of travel. A nozzle **2220-1** is attached to arm **2200-1** via a bracket **2210-1**. Optionally, a slot **2205-1** can be disposed in arm **2200-1** to permit adjustment of nozzle **2220-1** front to back along the direction of travel. Nozzle **2220-1** is connected to a fluid source (not shown) for applying fluid to the soil surface.

In an alternative embodiment as illustrated in FIG. **42** or FIG. **43**, a nozzle **2300** (**2300-2**) can be connected to one or both soil deflectors **3100**, **3500**. Nozzle **2300** is connected to a fluid source (not shown). While illustrated with nozzle **2300** depositing fluid rearward of soil deflector **3100** or **3500** opposite the direction of travel, nozzle **2300** can be positioned to apply fluid towards the trench, or nozzle **2300** can be positioned to deposit fluid away from the trench.

FIGS. **44** to **54** illustrate a scraper **3600** (**3600-1**, **3600-2**) for press wheel **255**. As illustrated in these figures, press wheel **255E** is the press wheel described in U.S. Application No. 62/843,037, filed 3 May 2019. While illustrated with this press wheel **255E**, scraper **3600** can be used with any of the other press wheels **255** described above.

As seen in FIG. **47**, scraper **3600** (**3600-1**, **3600-2**) is connected to axle **258** (**258-1**, **258-2**). In an alternative embodiment, scraper **3600** (**3600-1**, **3600-2**) can be connected to axle hub **1575** (**1575-1**, **1575-2**) or mounting arm **1570** (not shown). As seen in FIG. **54**, scraper **3600** (**3600-1**, **3600-2**) is disposed at inner side **3650** (**3650-1**, **3650-2**) of wheel **255E** (**255E-1**, **255E-2**). Scraper **3600** (**3600-1**, **3600-2**) has a base **3601** (**3601-1**, **3601-2**) which is connected to axle **258** (**258-1**, **258-2**), axle hub **1575** (**1575-1**, **1575-2**), or mounting arm **1570** as described above. Connected to base **3601** **3601** (**3601-1**, **3601-2**) is a first end **3602** (**3602-1**, **3602-2**) that extends to a second end **3603** (**3603-1**, **3603-2**). In one embodiment, first end **3602** (**3602-1**, **3602-2**) is wider than second end **3603** (**3603-1**, **3603-2**) such that there is a taper from first end **3602** (**3602-1**, **3602-2**) that extends to a second end **3603** (**3603-1**, **3603-2**). In one embodiment, scraper **3600** (**3600-1**, **3600-2**) is oriented such that second

end **3603** (**3603-1**, **3603-2**) is further from the seed trench as compared to first end **3602** (**3602-1**, **3602-2**).

As viewed in FIG. 51, there is an angle α ($\alpha-1$, $\alpha-2$) between base **3601** (**3601-1**, **3601-2**) and first end **3602** (**3602-1**, **3602-2**). In some embodiments, the angle is from 90° to 180° . In other embodiments, the angle is 120° - 175° , 140° - 170° , 150° - 160° , or about 155° .

FIG. 55 illustrates closing wheel **254E-2**. While illustrated for closing wheel **254E-2**, the same can apply to closing wheel **254E-1**. Closing wheel **254** can have one or more depth indicators **3700** (**3700-a**, **3700-b**). Depth indicators **3700** indicate the depth that closing wheel **254E** has penetrated the soil. This can provide a quick visual indication of the depth by looking at depth indicator **3700**. Depth indicator **3700** may or may not include markings for the actual depth associated with depth indicator **3700**. There can be any number of depth indicators **3700**. The number of depth indicators **3700** can be limited to a number for different selected depths. For example, if typical depth will be 1.5" or 1.75", there will be just two depth indicators **3700**. Depth indicators **3700** can be disposed as a circle around closing wheel **254**. The placement of depth indicator **3700** is based on a vertical distance that is perpendicular to the plane that the furthest point on closing wheel **254E** touches on the plane. In the illustrated embodiments, closing wheel **254E** is curved. The placement of depth indicator **3700** on closing wheel **254E** accounts for the curve as is it is placed as described previously.

FIGS. 56 to 75 illustrate a scraper **4600** (**4600-1**, **4600-2**) for press wheel **255E** (**255E-1**, **255E-2**). Scraper **4600** can be used with any of the press wheels **255** described above.

As seen in FIGS. 56-57, scraper **4600** (**4600-1**, **4600-2**) is connected to axle **258** (**258-1**, **258-2**). In an alternative embodiment, scraper **4600** (**4600-1**, **4600-2**) can be connected to axle hub **1575** (**1575-1**, **1575-2**) or mounting arm **1570** (not shown). As seen in FIG. 54, scraper **4600** (**4600-1**, **4600-2**) is disposed at inner side **3650** (**3650-1**, **3650-2**) of wheel **255E** (**255E-1**, **255E-2**). Scraper **4600** (**4600-1**, **4600-2**) has a base **4601** (**4601-1**, **4601-2**) which is connected to axle **258** (**258-1**, **258-2**), axle hub **1575** (**1575-1**, **1575-2**), or mounting arm **1570** as described above. Connected to base **4601** **4601** (**4601-1**, **4601-2**) is a first end **4602** (**4602-1**, **4602-2**) that extends to a second end **4603** (**4603-1**, **4603-2**). In one embodiment, first end **4602** (**4602-1**, **4602-2**) is wider than second end **4603** (**4603-1**, **4603-2**) such that there is a taper from first end **4602** (**4602-1**, **4602-2**) that extends to a second end **4603** (**4603-1**, **4603-2**). In one embodiment, scraper **4600** (**4600-1**, **4600-2**) is oriented such that second end **4603** (**4603-1**, **4603-2**) is further from the seed trench as compared to first end **4602** (**4602-1**, **4602-2**). In another embodiment, first end **4602** (**4602-1**, **4602-2**) is generally perpendicular to base **4601** (**4601-1**, **4601-2**) having an angle of 80° to 100° or about 90° . In one embodiment, one scraper **4600-1** can be disposed with respect to the other scraper **4600-2** such that first end **4602-1** of right scraper **4600-1** is disposed vertically above and at least partially overlaps first end **4602-2** of left scraper **4600-2**. This allows both scrapers **4600-1**, **4600-2** to fit between press wheels **255-1**, **255-2**. Second end **4603** (**4603-1**, **4603-2**) can be shaped to conform to inner side **3650** (**3650-1**, **3650-2**) of press wheel **255E** (**255E-1**, **255E-2**).

EXAMPLES

The following are non-limiting examples of various embodiments.

Example 1—A trench closing system for a row unit of an agricultural planter, the row unit having a row unit frame supporting an opener disk for opening a seed trench in a soil surface as the row unit travels in a forward direction of travel, the trench closing system comprising: a main frame supported by and extending rearwardly from the row unit frame; a frame member pivotally supported from the main frame; a pair of closing wheels each rotatably supported by the frame member with a first one of the pair of closing wheels disposed on a first side of the seed trench and a second one of the pair of closing wheels disposed on a second side of the seed trench, the pair of closing wheels cooperating with one another to close the opened seed trench with soil as the row unit travels in the forward direction of travel; a mounting arm pivotally mounted to said frame member, said mounting arm supporting a press wheel rearward of said closing wheels, said press wheel having an inner side disposed proximate to said frame member; a scraper disposed adjacent to said inner side of said press wheel.

Example 2.—the trench closing system of Example 1, wherein said scraper is connect to an axle that connects said press wheel to said frame member.

Example 3—the trench closing system of Example 1, wherein said scraper has a base for connecting to said frame member, a first end connected to said base, and a second end extending from said first end.

Example 4—the trench closing system of Example 3, wherein said first end is wider than said second end.

Example 5—the trench closing system of Example 3, wherein an angle between said base and said first end is 90° to 180° .

Example 6—the trench closing system of Example 3, wherein an angle between said base and said first end is 120° - 175° .

Example 7—the trench closing system of Example 3, wherein an angle between said base and said first end is 140° - 170° .

Example 8—the trench closing system of Example 3, wherein an angle between said base and said first end is 150° - 160° .

Example 9—the trench closing system of Example 3, wherein an angle between said base and said first end is about 155° .

Example 10—the trench closing system of any of Examples 3 to 9, wherein said scraper is disposed such that said second end is further from said seed trench than said first end.

Example 11—the trench closing system of any of any preceding Example further comprising a second press wheel supported by said mounting arm and a second scraper disposed adjacent to an inner side of said second press wheel.

Example 12—the trench closing system of Example 11, wherein the scraper is disposed above and partially overlaps the second scraper.

Various embodiments of the invention have been described above for purposes of illustrating the details thereof and to enable one of ordinary skill in the art to make and use the invention. The details and features of the disclosed embodiments are not intended to be limiting, as many variations and modifications will be readily apparent to those of skill in the art.

The invention claimed is:

1. A trench closing system for a row unit of an agricultural planter, the row unit having a row unit frame supporting an

11

opener disk for opening a seed trench in a soil surface as the row unit travels in a forward direction of travel, the trench closing system comprising:

- a main frame supported by and extending rearwardly from the row unit frame;
 - a frame member pivotally supported from the main frame;
 - a pair of closing wheels each rotatably supported by the frame member with a first one of the pair of closing wheels disposed on a first side of the seed trench and a second one of the pair of closing wheels disposed on a second side of the seed trench, the pair of closing wheels cooperating with one another to close the opened seed trench with soil as the row unit travels in the forward direction of travel;
 - a mounting arm pivotally mounted to said frame member, said mounting arm supporting a press wheel rearward of said closing wheels, said press wheel having an inner side disposed proximate to said frame member;
 - a scraper disposed adjacent to said inner side of said press wheel;
 - wherein said scraper has a base for connecting to said frame member, a first end connected to said base, and a second end extending from said first end; and
 - wherein said scraper is disposed such that said second end is further from said seed trench than said first end.
2. The trench closing system of claim 1, wherein said scraper is connected to an axle that connects said press wheel to said frame member.
3. The trench closing system of claim 1, wherein said scraper has a base for connecting to said frame member, a first end connected to said base, and a second end extending from said first end.
4. The trench closing system of claim 3, wherein said first end is wider than said second end.
5. A trench closing system for a row unit of an agricultural planter, the row unit having a row unit frame supporting an opener disk for opening a seed trench in a soil surface as the row unit travels in a forward direction of travel, the trench closing system comprising:
- a main frame supported by and extending rearwardly from the row unit frame;
 - a frame member pivotally supported from the main frame;
 - a pair of closing wheels each rotatably supported by the frame member with a first one of the pair of closing wheels disposed on a first side of the seed trench and a second one of the pair of closing wheels disposed on a second side of the seed trench, the pair of closing wheels cooperating with one another to close the

12

- opened seed trench with soil as the row unit travels in the forward direction of travel;
 - a mounting arm pivotally mounted to said frame member, said mounting arm supporting a press wheel rearward of said closing wheels, said press wheel having an inner side disposed proximate to said frame member;
 - a scraper disposed adjacent to said inner side of said press wheel;
 - wherein said scraper has a base for connecting to said frame member, a first end connected to said base, and a second end extending from said first end; and
 - wherein an angle between said base and said first end is 120°-175°.
6. The trench closing system of claim 5, wherein an angle between said base and said first end is 140°-170°.
7. The trench closing system of claim 5, wherein an angle between said base and said first end is 150°-160°.
8. The trench closing system of claim 5, wherein an angle between said base and said first end is about 155°.
9. A trench closing system for a row unit of an agricultural planter, the row unit having a row unit frame supporting an opener disk for opening a seed trench in a soil surface as the row unit travels in a forward direction of travel, the trench closing system comprising:
- a main frame supported by and extending rearwardly from the row unit frame;
 - a frame member pivotally supported from the main frame;
 - a pair of closing wheels each rotatably supported by the frame member with a first one of the pair of closing wheels disposed on a first side of the seed trench and a second one of the pair of closing wheels disposed on a second side of the seed trench, the pair of closing wheels cooperating with one another to close the opened seed trench with soil as the row unit travels in the forward direction of travel;
 - a mounting arm pivotally mounted to said frame member, said mounting arm supporting a press wheel rearward of said closing wheels, said press wheel having an inner side disposed proximate to said frame member;
 - a scraper disposed adjacent to said inner side of said press wheel;
 - a second press wheel supported by said mounting arm and a second scraper disposed adjacent to an inner side of said second press wheel; and
 - wherein the scraper is disposed above and at least partially overlaps the second scraper.

* * * * *